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Increasing Rigor and Accessibility of Young's Scoring Rule Proof

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Ich versichere hiermit, dass ich die von mir eingereichte Abschlussarbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

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Thank you!

Abstract

In 1975, H.P. Young stated that a social choice function is a composed scoring rule if and only if it is symmetric and consistent [You75]. The main purpose of this article is to improve rigor and accessibility of the original proof by Young. To begin with, we give a short introduction to the theory of voting. We formally define social choice functions, scoring functions, symmetry, and consistency. After that, we extend the domain of a symmetric and consistent social choice function from $\mathbb{N}^{m!}$ to $\mathbb{Q}^{m!}$. Then, we recall some results from convex analysis. Next, we give an illustrative proof of a special case of Young's characterization and present some visualizations. Afterward, we prove the dependency lemma, which is crucial for Young's proof, both formally and intuitively, with illustrations. Ultimately, we use the dependency lemma to restrict the domain of symmetric and consistent social choice functions and prove Young's characterization. On top of that, we give visualizations for the proof of composed scoring rules.

Zusammenfassung

Im Jahr 1975 stellt H.P. Young fest, dass eine Sozialwahlfunktion genau dann eine zusammengesetzte Scoring Regel ist, wenn die symmetrisch und konsistent ist [You75]. Der Hauptzweck dieses Artikels besteht darin, die Genauigkeit und die Zugänglichkeit des originalen Beweises von Young zu verbessern. Zunächst geben wir eine kurze Einführung zur Theorie der Wahlen. Wir definieren Sozialwahlfunktionen, Scoring-Regeln, Symmetrie und Konsistenz. Danach erweitern wir den Definitionsbereich einer symmetrischen und konsistenten Sozialwahlfunktion von $\mathbb{N}^{m!}$ zu $\mathbb{Q}^{m!}$. Anschließend erinnnern wir uns an einige Resultaten aus der konvexen Analyis. Danach geben wir einen illustrativen Beweis für einen Spezialfall von Young's Charakterizierung und präsentieren wir ein paar visualisierungen. Nachher beweisen wir das Abhängigkeitslemma, das entscheidend is für Young's Beweis, gleichzeitig formal und auch noch intuitiv mit Bildern. Letztlich benutzen wir die Abhängigkeitslemma, um den Definitionsbereich einer symmetrischen und konsistenten Sozialwahlfunktion zu beschränken und beweisen wir Young's Charakterisierung. Darüber hinaus geben wir Visualisierungen zum Beweis über zusammengesetzten Scoring-Regeln.

Contents

1 Introduction

Voting is relevant for group decision-making and is thus studied in social choice theory. There are different kinds of voting rules (or social choice functions), such as plurality, Borda, approval, or Condorcet voting. On the other hand, there are different kinds of axioms used to characterize them. Some impossibility theorems, such as the ones of Arrow [Arr63] or Gibbard-Satterthwaite [Gib73; Sat75], suggest that no voting rule is perfect in the sense that they all fail to satisfy some of the axioms at the same time. Therefore, the mechanism designers are forced to decide which axioms are more relevant for them. This leads to the so-called "axiomatic characterizations", which states that if the designer considers a certain set of axioms to be important, then only a small class of voting rules (or only a single rule) can be chosen.

H. P. Young, in 1975, using some of the axioms, gave a characterization of scoring rules [You75], which is an important class of social choice functions. Young claims that a social choice function is a composed scoring rule if and only if it satisfies the following two axioms:

1. It is symmetric, which means that all voters and alternatives are treated equally.
2. It is consistent, which means that the alternatives winning in two different elections will again be chosen if the two groups of voters are merged.

Scoring rules are the social choice functions, which choose the alternative with the highest score. The scores of the alternatives are obtained by observing how many times one alternative is ranked on each position, and each position is assigned a score. Composed scoring rules are obtained by implementing scoring rules in multiple rounds, with different scores.

Unlike symmetry, which is almost a basic requirement for social choice functions — the real characterizing property is consistency. There have been plenty of research done regarding consistency in different settings. Brandl and Peter, in 2022, characterized approval voting using consistency (see [BP22]). In 2016, Brandl, Brandt, and Seedig studied the consistency axiom in a probabilitic setting (probabilistic social choice)(see [BBS16]). In 1978, Young and Levenglick showed that consistency is compatible with Condorcet's principle (see [YL78]). Two recent papers by Dong and Lederer in 2023 and 2024 investigated the consistency axiom under the approval-based committee (ABC) setting (see [DL23; DL24]). Moreover, Myerson, in 1995, and Pivato, in 2013, presented an alternative proof of Young's characterization using different techniques (see [Mye95; Piv13]).

The original proof [You75] by Young is quite technical and tough to read. In this article, we aim to improve its rigor and accessibility by giving detailed explanations, more examples, and some visualizations.

1.1 Setting and Definitions

Let's take a look at the general setting and some definitions.

First, we need to have a set of voters. This could be $\mathbb{N}$, the set of non-negative integers (however, only finite number of voters can vote). Then, we need to have an alternative set $A = \{a_1, a_2, \ldots, a_m\}$ to choose from. With m alternatives, there are $m!$ possible orders. Let $\mathscr{P}$ denote the set of $m!$ distinct preference orders on the fixed alternative set A. For any finite set of voters $V \subset \mathbb{N}$, a profile is a function from V to $\mathscr{P}$. So the profile collects individual decisions. And an SCF (Social Choice Function) is a function from the set X of all profiles to the family of non-empty subsets of A, so it is the function which collects individual preferences and makes group decisions. Namely, the SCF selects an alternative according to the group's profile.

A SCF is said to be anonymous if it depends only on the number of voters associated with each preference order. In this case, we can represent the domain of such anonymous SCFs by $\mathbb{N}^{m!}$. For any $x \in \mathbb{N}^{m!}$ and $p \in \mathscr{P}$, x_p represents the number of voters having preference order p. Notice that this domain only makes sense because of anonymity.

Let $\sigma \in S_m$, where S_m is the permutation group on $\{1, 2, \ldots, m\}$, be a permutation of the alternatives. We say a SCF f is neutral if $f \circ \sigma = \sigma \circ f$ for all $\sigma \in S_m$. This means that renaming the alternatives should not affect the outcome. If f is both anonymous and neutral, it is said to be symmetric. Symmetric means that all voters and all alternatives are treated equally. Note that symmetry is a very natural requirement for most group decision situations. In fact, they are often minimal requirements for SCFs. There is little cost in requiring them, and plenty of SCFs satisfy them (see [Bra+16], sec 2.3).

Let M_σ denote a permutation matrix with dimension $m! \times m!$ (see Example 1.2). For a symmetric and anonymous SCF f, it holds that:

$$f(M_\sigma(x)) = \sigma(f(x)) \quad \text{for all } x \in \mathbb{N}^{m!}. \tag{1}$$

Example 1.1. *Assume $A = \{a, b, c\}$. Then there are $3! = 6$ distinct preference orders, with $p_1 = (a, b, c)^T$, $p_2 = (a, c, b)^T$, $p_3 = (b, a, c)^T$, $p_4 = (b, c, a)^T$, $p_5 = (c, a, b)^T$, $p_6 = (c, b, a)^T$. Then $\mathscr{P} = \{p_1, p_2, \ldots, p_6\}$, is the set of distinct preference orders. Let $V = \{1, 2, 3, 4, 5\}$ be a set of voters. Let the profile be defined as:*

$$P : V \to \mathscr{P}, \quad 1 \mapsto p_1,\ 2 \mapsto p_3,\ 3 \mapsto p_3,\ 4 \mapsto p_3,\ 5 \mapsto p_4$$

X is the set of all possible profiles. SCF is $f : X \to \mathcal{P}(A) \setminus \varnothing$. (We use $\mathcal{P}(A)$ to denote the power set of A.) Assume f to be anonymous, then we can represent X as $\mathbb{N}^{3!} = \mathbb{N}^6$. Thus, according to P, we have $x = (1, 0, 3, 1, 0, 0)^T \in X = \mathbb{N}^6$ and x_i represents the number of voters having preference order p_i. With f = Plurality we get $f(x) = \{b\}$. (Plurality rule chooses the alternative being ranked the most in the first place.)

Example 1.2. *Considering again the setting in Example 1.1. Let $\sigma = \begin{pmatrix} a & b & c \\ b & a & c \end{pmatrix}$ be a permutation of the alternatives. Applying the permutation yields $p'_1 = (b, a, c)^T$, $p'_2 = (b, c, a)^T$, $p'_3 = (a, b, c)^T$, $p'_4 = (a, c, b)^T$, $p'_5 = (c, b, a)^T$, $p'_6 = (c, a, b)^T$. The profile is permutated in the* *following way:*

$$\sigma_{\mathscr{P}} = \begin{pmatrix} p_1 & p_2 & p_3 & p_4 & p_5 & p_6 \\ p_3 & p_4 & p_1 & p_2 & p_6 & p_5 \end{pmatrix}.$$

So the deduced M_σ should be

$$M_\sigma = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

We use again $x = (1, 0, 3, 1, 0, 0)^T$. Notice the profile is now:

$$P' : V \to \mathscr{P}, \quad 1 \mapsto p_3,\ 2 \mapsto p_1,\ 3 \mapsto p_1,\ 4 \mapsto p_1,\ 5 \mapsto p_2.$$

So we have $x' = (3, 1, 1, 0, 0, 0)^T = M_\sigma(x)$ and $f(M_\sigma(x)) = \{a\} = \sigma(f(x))$, for f = Plurality, as expected.

Now we want to introduce another axiom, which is, in fact, the characterizing property of scoring rules, which will be proven later. Consider a voter group V' with its choice set A' and another disjoint voter group V'' with its choice set A''. Whenever $A' \cap A'' \neq \varnothing$, the group $V' \cup V''$ jointly should choose exactly the set $A' \cap A''$. Any SCF with the above property is said to be consistent. If f is anonymous, then the consistency condition can be formulated as follows:

$$f(x' + x'') = f(x') \cap f(x'') \quad \text{whenever } f(x') \cap f(x'') \neq \varnothing. \tag{2}$$

Notice the consistency axiom ensures that the alternatives chosen by the merged group are at least as good as any other alternatives, in the sense that these will be chosen by all the individual groups. The other way around, the alternatives not chosen by the merged group are not as good as others since they are excluded by at least one of the individual groups.

Example 1.3. *Consider again the setting from Example 1.1. Let $x' = (1, 0, 3, 1, 0, 0)^T$, $x'' = (1, 0, 1, 0, 0, 0)^T$. Using the plurality rule, we get $f(x') = \{b\}$ and $f(x'') = \{a, b\}$. Notice that $f(x' + x'') = f(2, 0, 4, 1, 0, 0)^T = \{b\} = f(x') \cap f(x'')$. This always holds because plurality is a scoring rule, and the latter one is consistent.*

Lastly, we introduce the continuity axiom. An anonymous f is said to be continuous if, whenever $f(x) = \{a_i\}$, then for any profile y there is a sufficiently large integer n such that $f(y + n'x) = \{a_i\}$ for all $n' \geq n$. This axiom ensures that if there are two disjoint committees, then replicate one committee a sufficient number of times, so this committee will eventually dominate the election.

1.2 Scoring Rules

Next, we introduce the scoring functions. Given m alternatives and a profile, assign a score of $s_i \in \mathbb{R}$ to each voter's i-th most preferred alternative, and let the choice set consist of the alternatives with the highest total score. Any SCF obtained this way will be called a (simple) scoring function, or (simple) scoring rule, denoted by f^s, where s, called a scoring vector, is the m-vector $(s_1, s_2, \ldots, s_m)^T \in \mathbb{R}^m$.

A well-known scoring rule is Borda's rule. For m alternatives, we assign a score $m-1$ to each voter's most preferred alternative, a score $m-2$ to their second preferred, and so on. The least preferred alternative will be assigned a zero score. Another common scoring rule is the plurality rule, with scoring vector $(1, 0, \ldots, 0)^T \in \mathbb{R}^m$.

Now we want to define the scoring rules formally:

Definition 1.4 (Scoring rules). *Given $p \in \mathscr{P}$, let E_p be the $m \times m$ permutation matrix with "1" in the position (i, j) if and only if a_i is j-th most preferred in the preference order p. Then, for every $x \in \mathbb{R}^{m!}$, define $D(x) = \sum_{p \in \mathscr{P}} x_p E_p \in \mathbb{R}^{m \times m}$. Let $D_i(x)$ denote the i-th row of $D(x)$. We say f is a scoring rule if the following is satisfied:*

$$a_i \in f^s \quad \text{if and only if} \quad D_i(x) \cdot s \geq D_j(x) \cdot s \quad \text{for all } j, \quad 1 \leq j \leq m. \tag{3}$$

Example 1.5. *Considering again the setting in example 1.1. We have*

$$E_{p_1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \; E_{p_2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \; E_{p_3} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$
$$E_{p_4} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \; E_{p_5} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \; E_{p_6} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Let's take E_{p_5} as an example. Recall that $p_5 = (c, a, b)^T$. So the first alternative a is ranked in the second place, that is why we have $(0, 1, 0)$ in the first row. Similarly, we get the second and third rows by finding out how b and c are ranked.

Let $x = (1, 0, 3, 1, 0, 0)^T$, then

$$\begin{aligned} D(x) &= \sum_{p \in \mathscr{P}} x_p E_p = x_{p_1} E_{p_1} + \cdots + x_{p_6} E_{p_6} \\ &= 1 \cdot E_{p_1} + 3 \cdot E_{p_3} + 1 \cdot E_{p_4} \\ &= \begin{pmatrix} 1 & 3 & 1 \\ 4 & 1 & 0 \\ 0 & 1 & 4 \end{pmatrix}. \end{aligned}$$

If $f = \text{Plurality}$, then $s = (1, 0, 0)^T$. We can calculate that $D_1(x) \cdot s = 1,\ D_2(x) \cdot s = 4,\ D_3(x) \cdot s = 0$. It follows that $f^s(x) = \{b\}$.

Note that $D_{ij}(x)$ is how many times the alternative i is ranked in the j-th place for profile x. Note also that the sum of each row and each column is the number of voters, thus always the same.

The trivial SCF is that f such that $f(x) = A$ for all profiles x. The trivial SCF can be represented by the simple scoring function $f^{(0,0,\ldots,0)}$.

If f and g are SCFs such that $f(x) \subseteq g(x)$ for all profiles x, we say that f is a refinement of g and write $f \precsim g$ ($f \prec g$ if $f \precsim g$ and $f \neq g$). For any anonymous SCF g and scoring vector $s \in \mathbb{R}^m$, the composition of f^s with g, denoted by $f^s \circ g$, is defined by

$$a_i \in f^s \circ g(x) \quad \text{if and only if} \quad a_i \in g(x) \text{ and } D_i(x) \cdot s \geq D_j(x) \cdot s, \tag{4}$$

whenever $a_j \in g(x)$ (notice that we always calculate with all alternatives without deleting).

The meaning of $f^s \circ g$ is that the function f^s is used to resolve ties produced by g. Applying (4) recursively, we define a composed scoring rule to be any SCF g of the form $f^{s^\alpha} \circ f^{s^{\alpha-1}} \circ \cdots \circ f^{s^1}$, $\alpha \geq 1$, where $s^1, s^2, \ldots, s^\alpha \in \mathbb{R}^m$. The expression $f^{s^\alpha} \circ f^{s^{\alpha-1}} \circ \cdots \circ f^{s^1}$ is called a composition series for g of length α.

Example 1.6. *The composed scoring rule $f^{(0,1,0)} \circ f^{(1,0,0)}$ is not continuous. Let us consider again the setting in example 1.1: Assume $A = \{a, b, c\}$, with $p_1 = (a, b, c)^T$, $p_2 = (a, c, b)^T$, $p_3 = (b, a, c)^T$, $p_4 = (b, c, a)^T$, $p_5 = (c, a, b)^T$, $p_6 = (c, b, a)^T$. Let $x = (0, 1, 0, 0, 0, 1)^T$ be a profile. Then*

$$f^{(0,1,0)} \circ f^{(1,0,0)}(x) = \{c\}.$$

In the first round $f^{(1,0,0)}(x)$ chooses a and c. In the second round c wins. Let $y = (0, 1, 0, 0, 0, 0)^T$ be another profile. Then clearly

$$f^{(0,1,0)} \circ f^{(1,0,0)}(y + nx) = f^{(0,1,0)} \circ f^{(1,0,0)}((0, 1+n, 0, 0, 0, n)^T) = \{a\} \quad \forall n \in \mathbb{N}.$$

So multiplying the profile x can not affect the result. Hence the composed scoring rule is not necessarily continuous.

1.3 Extension of Domain

We now show how to extend f in a natural way from the domain $\mathbb{N}^{m!}$ to $\mathbb{Q}^{m!}$. In other words, we will allow profiles to have fractional and negative numbers of voters. This is technically important for the later proof, because we will in fact prove Young's characterization on $\mathbb{Q}^{m!}$. We assume f to be symmetric and consistent.

Let $e \in \mathbb{N}^{m!}$ be the vector with "1" in every component. By symmetry, $f(ne) = A$ for all $n \in \mathbb{N}$. Now define $f(x - ne) := f(x)$ for each $n \in \mathbb{N}$. We want to show this is well-defined: Let $x' - n'e = x - ne$, without loss of generality, let $n' > n$, then $x' \neq x$. We need to show that, with this definition, we still get $f(x' - n'e) := f(x') \overset{(*)}{=} f(x) =: f(x - ne)$. To prove $(*)$, we need to use consistency:

$$f(x') \overset{\text{given}}{=} f(x - ne + n'e) = f(x + (n'-n)e)$$
$$\overset{\text{consistency}}{=} f(x) \cap f((n'-n)e) = f(x) \cap A = f(x).$$

This extends f to $\mathbb{Z}^{m!}$ and it is the unique extension of f to $\mathbb{Z}^{m!}$ that is symmetric and consistent on $\mathbb{Z}^{m!}$. For any positive integer n and $x \in \mathbb{Z}^{m!}$, consistency implies that

$$f(nx) = f(\underbrace{x + \cdots + x}_{\text{n times}}) \overset{\text{consistency}}{=} \underbrace{f(x) \cap \cdots \cap f(x)}_{\text{n times}} = f(x),$$

that is, f is homogeneous. Now for each positive integer n, we can define $f(\frac{x}{n}) = f(x)$. This is again well-defined. Because similarly, if $\frac{x'}{n'} = \frac{x}{n}$, then

$$f\left(\frac{x'}{n'}\right) := f(x') \overset{\text{homogeneous}}{=} f(nx') \overset{\text{given}}{=} f(n'x) \overset{\text{homogeneous}}{=} f(x) =: f\left(\frac{x}{n}\right).$$

This extends f to $\mathbb{Q}^{m!}$ and it is the unique such extension that is symmetric and consistent on $\mathbb{Q}^{m!}$.

1.4 Convexity

In the later proof, we will make use of that the preimages are convex. Thus, we would give an introduction to convexity here.

Let f be a symmetric and consistent SCF. Consider the domain of f to be $\mathbb{Q}^{m!}$. For each i, $1 \le i \le m$, let $R_i = \{x \in \mathbb{Q}^{m!} : a_i \in f(x)\}$. For any $x, y \in R_i$ and rational λ such that $0 \le \lambda \le 1$, we have $a_i \in f(\lambda x)$ and $a_i \in f((1-\lambda)x)$, so by consistency $a_i \in f(\lambda x + (1-\lambda)y)$, that is, $\lambda x + (1-\lambda)y \in R_i$. Thus, each R_i is $\mathbb{Q}$-convex. In fact, R_i is a $\mathbb{Q}$-convex cone, since $x \in R_i$ implies $\lambda x \in R_i$ for all rational $\lambda > 0$.

For $S \subseteq \mathbb{R}^n$, let $\mathbf{cvx}\, S$ denote the convex hull of S, $\mathbf{aff}\, S$ the affine hull of S, and $\overline{S}$ the closure of S. If $S \subseteq W \subseteq \mathbb{R}^n$, where W is an affine set, let $\mathbf{int}_W\, S$ denote the interior of S relative to W, and $\mathbf{ri}\, S := \mathbf{int}_{\mathbf{aff}\, S}\, S$ the relative interior of S. Let $\mathbf{dim}\, S$ denote the dimension of $\mathbf{aff}\, S$.

The reason of introducing both $\mathbb{Q}$-convex and $\mathbb{R}$-convex is that the SCFs are defined on $\mathbb{Q}$; however, the hyperplane argument, which is essential for the proof, only works in $\mathbb{R}$.

We want to simply state some technical results about $\mathbb{Q}$- and $\mathbb{R}$-convexity. Notice if not specified, “convex” means $\mathbb{R}$-convex.

Lemma 1.7. *If C is convex, then $\overline{C}$ and $\mathbf{ri}\, C$ are convex and $\mathbf{ri}\, \overline{C} = \mathbf{ri}\, C$, $\overline{\mathbf{ri}\, C} = \overline{C}$.*

Proof. This is a known fact. A proof can be found in e.g. sec 14 from [Ulb23]. □

Lemma 1.8. *$C \subseteq \mathbb{Q}^n$ is $\mathbb{Q}$-convex if and only if $C = \mathbb{Q}^n \cap \mathbf{cvx}\, C$.*

Proof. See [You75]. □

Lemma 1.9. *If $C \subseteq \mathbb{Q}^n$ is $\mathbb{Q}$-convex, then $\overline{C}$ is convex.*

Proof. See [You75]. □

Lemma 1.10. *If $C = \bigcup_{i=1}^{k} S_i$, where $C \subseteq \mathbb{R}^n$ is convex and k is finite, then for some i,* $\mathbf{dim}\, C = \mathbf{dim}\, S_i$.

Proof. See [You75]. □

Lemma 1.7 and Lemma 1.8 are used in Section 3.2.3 to derive a contradiction. Using this contradiction, we immediately after showed that the interior of preimage-cones R_i do not intersect. Lemma 1.9 and Lemma 1.10 are used in Section 3.2.2 to show the convexity of $\overline{R_i}$, and further the interior of $\overline{R_i}$ is not empty. For intuition of preimage-cones, see Figure 4.

2 Some Intuition

Before starting the formal proof, it is worth getting some intuition first. In this section, we will therefore consider cases with two and three alternatives.

2.1 Proof of Two Alternatives

Let's consider a simple case with only two alternatives $A = \{a, b\}$. Let f be a consistent, symmetric SCF. We want to show that f is a scoring rule. (Notice, in general, with symmetry and consistency, we can only show this is a composed scoring rule. But in the case of 2 alternatives, we can show more. Namely, it is a simple scoring rule.)

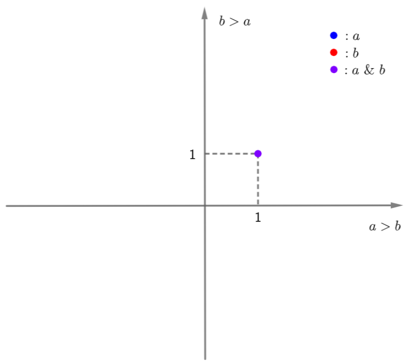
In this case, there are only two possible preferences. Thus the domain of f is simply two-dimensional. Notice that we have already extended the domain to $\mathbb{Q}$ in Section 1.3. So every point from $\mathbb{Q}^2$ will be mapped to either $\{a\}$, $\{b\}$ or $\{a, b\}$ by f. As in Figure 1, we color every rational point either blue, red or violet for f chooses $\{a\}$, $\{b\}$ or $\{a, b\}$. Since f is symmetric, using $\sigma = (1\ 2)$, it follows that

$$f(y, x) = \sigma(f(x, y)). \tag{5}$$

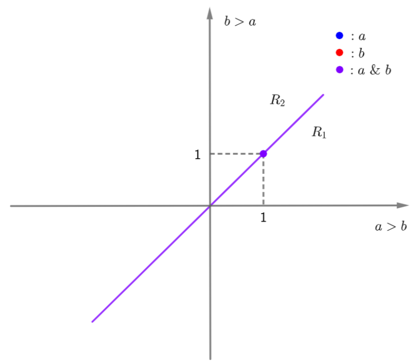
Consider the point $(1, 1)$. If this point chooses a, using (5), we get $\{a\} = f(1, 1) = \{b\}$, a contradiction. For the same reason, this point can not choose $\{b\}$. Thus $f(1, 1) = \{a, b\}$ (Figure 1a). Recall that $f(\lambda x) = f(x)$ for $\lambda > 0$ using consistency, we have $f(\lambda x) = \{a, b\}$ for $\lambda > 0$. Recall $f(0) = f((1, 1) - e) = f((1, 1) - 2e) = f(1, 1) = f(-1, -1) = \{a, b\}$ by domain extension, we conclude that every rational point on the diagonal $y = x$ would be mapped to {a,b}, as Figure 1b shows. We denote the region under the diagonal with R_1 and the region above with R_2.

Next, assume there is another violet x not on the diagonal. Without loss of generality, assume such x exists in R_1. Then use again $f(\lambda x) = f(x)$ for $\lambda > 0$, we get Figure 2a. Recall that

$$f(\lambda x) + f((1 - \lambda)y) = f(x) \cap f(y) \quad \text{for} \ \ 0 \leq \lambda \leq 1, \tag{6}$$



(a) Step 1



(b) Step 2

Figure 1: Two alternatives

by simply using consistency and domain extension, we conclude that the whole R_1 must be violet. Then we use Equation (5) to deduce that R_2 must also be violet. Thus, in this case, the function will choose the entire alternative set A (Figure 3a). We know this is a trivial scoring rule with scoring vector $s = (0, 0)^T$.

Otherwise, there is no violet x, except these on the diagonal. Without loss of generality, assume there is a blue x in R_1. Then Equation (5) implies that there exists a red x' in R_2 (Figure 2b). Now, using the convexity property (6), we conclude that the whole R_1 would be blue, and the whole R_2 would be red (Figure 3b). In this case, we can show that every vector $s \in R_1$, that is, $s = (s_1, s_2)^T$ with $s_1 > s_2$, is a proper scoring vector, especially the normal vector of the separating line $s = (1, -1)^T$. To see this, consider the function D defined in Definition 1.4. We have

$$D(x, y) = \begin{pmatrix} x & y \\ y & x \end{pmatrix}, \; D_1 \cdot s = s_1 \cdot x + s_2 \cdot y, \; D_2 \cdot s = s_1 \cdot y + s_2 \cdot x. \tag{7}$$

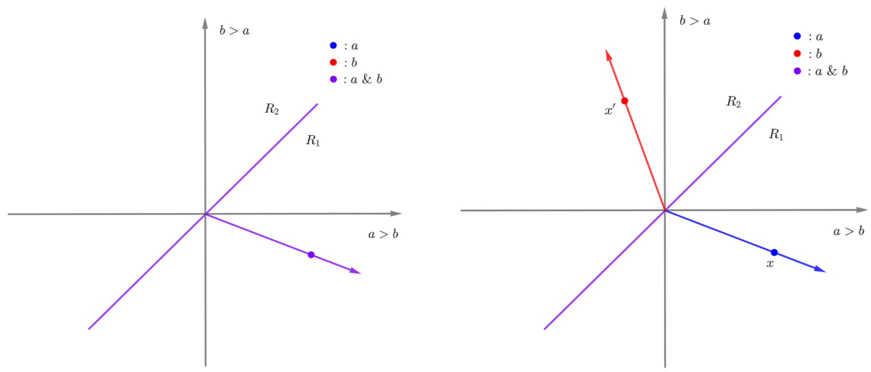
And we have

$$\begin{aligned} D_1 \cdot s > D_2 \cdot s \quad &\Leftrightarrow \quad s_1 \cdot x + s_2 \cdot y > s_1 \cdot y + s_2 \cdot x \\ &\Leftrightarrow \quad (s_1 - s_2) \cdot x > (s_1 - s_2) \cdot y \\ &\Leftrightarrow \quad x > y \qquad\qquad\qquad\qquad \text{since } s_1 > s_2 \end{aligned}$$

Notice that the points satisfying $x > y$ are the region R_1, which means a will be chosen exactly by the points in R_1. Similarly, b will be chosen alone if and only if $y > x$. So we conclude, in this case, f is also a scoring rule with scoring vector $s = (1, -1)^T$, for instance.

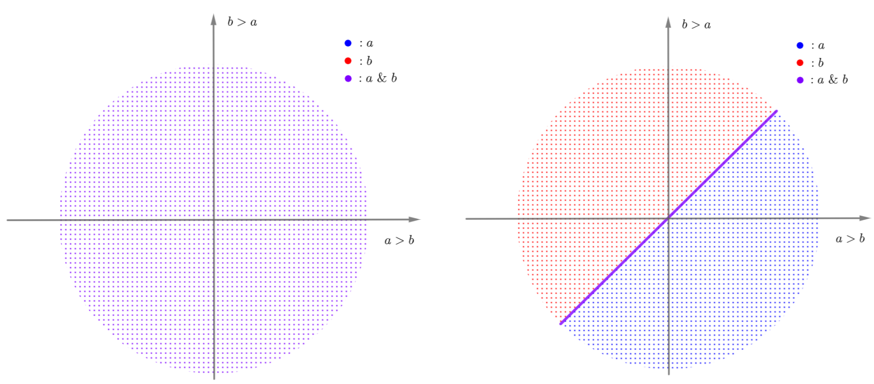
Notice that the scoring vector is not unique. We can always shift the scoring vector or multiply it with a positive number without changing the preimage of the function.

The last case is obtained by simply flipping the second case, and it would be also a scoring



(a) Case 1 (b) Case 2

Figure 2: Two alternatives



(a) Case 1 final (b) Case 2 final

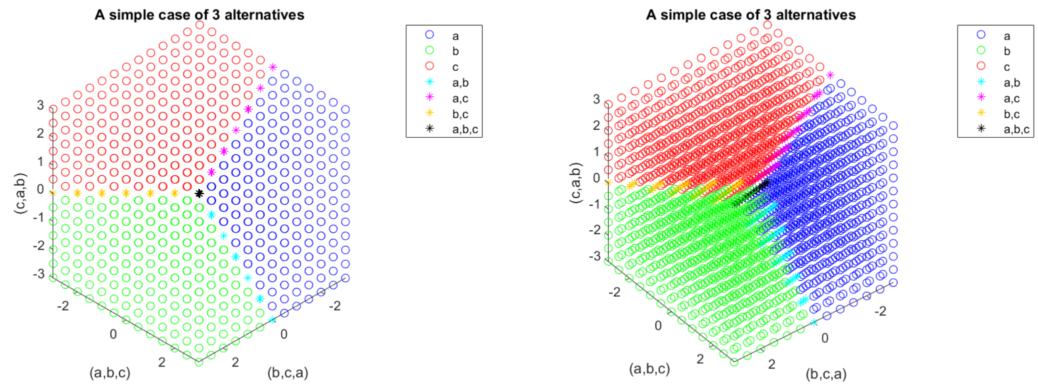
Figure 3: Two alternatives

rule with the opposite scoring vector $s = (-1, 1)^T$, for instance. So we see, in either of the three cases, f is a scoring rule by assuming f to be consistent and symmetric.

As we have seen, the case with two alternatives is simple and easy to visualize. However, if we consider the case with three alternatives, the domain is already 6-dimensional, which would be impossible to visualize. Nevertheless, we would like to give some intuition by considering a restricted situation.

2.2 Three Alternatives: A Restricted Case

Now, let us consider the case with three alternatives: $A = \{a, b, c\}$. As already mentioned, the real domain would have six dimensions. However, if we assume three of the components to be zero and only focus on the three non-zero components, we would get a nice visualization. In



(a) view 1

(b) view 2

Figure 4: Three alternatives (Section 2.2)

the following, let the three non-zero components correspond to preferences (a, b, c), (b, c, a) and (c, a, b), and we use Borda's rule. The figures (Figure 4) generated with MATLAB (source code in appendices) suggest the following (for simple scoring rules):

- The preimages are symmetric. The regions are convex cones.
- The points on the line $\lambda(1, 1, \ldots, 1)$ always choose all alternatives.
- The points on the boundary of two regions choose both alternatives, the points on the boundary of three regions choose all three alternatives, and so on.
- Dimension reduction with respect to the number of alternatives chosen. The regions choosing only one alternative have full dimensions. Regions choosing two alternatives have one less dimension. The region choosing all alternatives has only one dimension.

Notice that the two figures represent the same simple scoring rule but from different perspectives. In Figure 4a we have the vector $(-1, -1, -1)$ pointing towards us. With Figure 4b, we get some insight of the convex cone structure, which will be used in the proof later.

3 Proof of the Main Theorem

In this section, we will prove the main theorem. First, we state again the following scoring rule characterization by Young. Our main focus will be to improve its rigor and accessibility.

Theorem 3.1 (Young's Characterization, Young, 1975)**.**

(i.) A SCF is symmetric, consistent, and continuous if and only if it is a (simple) scoring rule.

(ii.) A SCF is symmetric and consistent if and only if it is a composed scoring rule.

The “if” direction is straightforward and will be proved right now.

3.1 Proof of the “if” Direction

In the following, we show that composed scoring rules are symmetric and consistent.

Proof. Let $A = \{a_1, \dots, a_m\}$ be a set of m-alternatives and $f = f^{s^\alpha} \circ \cdots \circ f^{s^1}$ be a composed scoring rule with a scoring vectors $s^j = (s^j_1, \dots, s^j_m)^T$. Let $x \in \mathbb{Q}^{m!}$ be a profile. Then according to the definition of the scoring function, the score for alternative a_i using f^{s^1} in round 1, denoted by Z^1_i, will be

$$Z^1_i = \sum_{k=1}^{m} s^1_k \cdot (\# \ a_i \text{ be ranked the k-th}).$$

According to the definition, the scoring function f^{s^1} chooses

$$f^{s^1}(x) = \{a_q : q \in \operatorname*{arg\,max}_{i \in \{1,\dots,m\}} Z^1_i\}.$$

for arbitrary profile x. If there is no tie, then we are finished. Otherwise, there are some tied round-1 winners, joining round-2. We denote this set by A^2 with $A^2 \subseteq A$. We then calculate

$$Z^2_i = \sum_{k=1}^{m} s^2_k \cdot (\# \ a_i \text{ be ranked the k-th}).$$

The round-2 winner will now be chosen:

$$f^{s^2} \circ f^{s^1}(x) = \{a_q : q \in \operatorname*{arg\,max}_{\substack{i \in \{1,\dots,m\} \\ a_i \in A^2}} Z^2_i\}.$$

Thus, we can define the round-j winner the same way, if the tie is not broken until round-$(j-1)$.

It is easy to see, during the process, only the number of each preference is involved. Therefore, the scoring function is anonymous. It is also obvious that the alternatives are treated equally: If the profiles are switched, the scores will be switched the same way, thus the result will also be switched the same way. Now, we want to show that the scoring function is consistent.

Let's say, $Z^1 = (Z^1_1, \dots, Z^1_m)$ is the scoring profile of round-1 for a group and $\tilde{Z}^1 = (\tilde{Z}^1_1, \dots, \tilde{Z}^1_m)$ the scoring profile of round-1 for another group. It is clear that

$$\operatorname*{arg\,max}_{i \in \{1,\dots,m\}} Z^1_i \bigcap \operatorname*{arg\,max}_{i \in \{1,\dots,m\}} \tilde{Z}^1_i = \operatorname*{arg\,max}_{i \in \{1,\dots,m\}} (Z^1_i + \tilde{Z}^1_i)$$

if

$$A^2 := \operatorname*{arg\,max}_{i \in \{1,\dots,m\}} Z^1_i \bigcap \operatorname*{arg\,max}_{i \in \{1,\dots,m\}} \tilde{Z}^1_i \neq \varnothing.$$

If there is no tie, then we are done. Otherwise, we notice that

$$\operatorname*{arg\,max}_{\substack{i \in \{1,\dots,m\} \\ a_i \in A^2}} Z^2_i \bigcap \operatorname*{arg\,max}_{\substack{i \in \{1,\dots,m\} \\ a_i \in A^2}} \tilde{Z}^2_i = \operatorname*{arg\,max}_{\substack{i \in \{1,\dots,m\} \\ a_i \in A^2}} (Z^2_i + \tilde{Z}^2_i)$$

if

$$A^3 := \underset{\substack{i\in\{1,\dots,m\}\\ a_i\in A^2}}{\arg\max}\, Z_i^2 \bigcap \underset{\substack{i\in\{1,\dots,m\}\\ a_i\in A^2}}{\arg\max}\, \tilde{Z}_i^2 \neq \varnothing.$$

Applying this recursively shows the consistency.

Next, we want to show, if f is a simple scoring rule, then f is also continuous. Let's assume that $\underset{i\in\{1,\dots,m\}}{\arg\max}\, Z_i^1 = d$. Then we notice that

$$\underset{i\in\{1,\dots,m\}}{\arg\max}\, (\tilde{Z}_i^1 + n \cdot Z_i^1) = d$$

for n big enough. This shows the continuity. But if f is a composed scoring rule, continuity does not always hold since example 1.6 serves as a counterexample. So we are done with the easy direction.

□

The difficult part is to prove the "only if" direction, that is, if a SCF is symmetric and consistent, then it is a composed scoring function. In order to understand the proof later, we reformulate the "only if" direction of Theorem 3.1 (i) formally:

Theorem 3.2 (Young's Characterization (i), "only if" direction, reformulated)**.** *Given $A = \{a_1, \dots, a_m\}$ and a function $f : \mathbb{Q}^{m!} \to \mathcal{P}(A) \setminus \varnothing$ which satisfies*

$$f(M_\sigma(x)) = \sigma(f(x)) \quad \textit{for all } x \in \mathbb{Q}^{m!}, M_\sigma \in \mathbb{Q}^{m!\times m!} \textit{ and} \tag{8}$$

$$f(x+y) = f(x) \cap f(y) \quad \textit{for all } x, y \in \mathbb{Q}^{m!} \textit{ that } f(x) \cap f(y) \neq \varnothing. \tag{9}$$

Define $D(x) = \sum_{p\in\mathcal{P}} x_p E_p \in \mathbb{Q}^{m\times m}$ as in Definition 1.4, then:

$$\begin{aligned} &\exists\ s \in \mathbb{Q}^m\ \forall x \in \mathbb{Q}^{m!} \quad \textit{it holds that:}\\ &a_i \in f(x) \ \Leftrightarrow\ \forall j:\ D_i(x)\cdot s \geq D_j(x)\cdot s. \end{aligned} \tag{10}$$

Observing the reformulated theorem above, we see that our overall plan is to make use of Equation (8) and Equation (9) to find the scoring vectors s in Equation (10).

3.2 Dependency Lemma

To begin with, we want to prove an important result, namely the dependency lemma, which shows that for symmetric and consistent SCF f with a profile x, $f(x)$ only depends on $D(x)$, with D defined earlier in Definition 1.4.

Lemma 3.3 (Dependency Lemma)**.** *Given $A = \{a_1, \dots, a_m\}$, $p \in \mathscr{P}$, let E_p be the $m \times m$ permutation matrix with "1" in the position (i, j) if and only if a_i is j-th most preferred in the preference order p. For every $x \in \mathbb{R}^{m!}$, define $D(x) := \sum_{p\in\mathscr{P}} x_p E_p \in \mathbb{R}^{m\times m}$. Then for any symmetric and consistent SCF f we have*

$$f(x) = A \quad \textit{for all } \ x \in R := \{x \in \mathbb{Q}^{m!} : D(x) = 0\}. \tag{11}$$

Before proving the dependency lemma, we firstly point out that D is linear, that is, $D(\lambda x) = \lambda D(x)$ and $D(x+y) = D(x) + D(y)$ for any profile $x, y \in \mathbb{Q}^{m!}$ and $\lambda \in \mathbb{R}$, simply because

$$\begin{aligned} D(\lambda x) &= \sum_{p\in\mathscr{P}} (\lambda x)_p E_p = \lambda \sum_{p\in\mathscr{P}} x_p E_p = \lambda D(x), \\ D(x+y) &= \sum_{p\in\mathscr{P}} (x+y)_p E_p = \sum_{p\in\mathscr{P}} x_p E_p + \sum_{p\in\mathscr{P}} y_p E_p = D(x) + D(y). \end{aligned}$$

3.2.1 Proof of Dependency Lemma for Three Alternatives

Next. we want to give an illustrative proof of the dependency lemma for three alternatives. Note that although the proof for three alternatives is simple, the general proof for m alternatives is not trivial.

Proof. Let f be symmetric and consistent. Considering the domain $\mathbb{Q}^6$ and let the six components represent preferences $(abc), (acb), (bac), (bca), (cab), (cba)$ respectively. With D defined in Section 1.2, we have

$$D(x_1, x_2, x_3, x_4, x_5, x_6) = \begin{pmatrix} x_1 + x_2 & x_3 + x_5 & x_4 + x_6 \\ x_3 + x_4 & x_1 + x_6 & x_2 + x_5 \\ x_5 + x_6 & x_2 + x_4 & x_1 + x_3 \end{pmatrix}.$$

If $D(y) = 0$, then $y = \lambda(1, -1, -1, 1, 1, -1)$, $\lambda \in \mathbb{Q}$, by simply solving the system of linear equations. If $\lambda > 0$, we have

$$\begin{aligned} f(y) &= f(1, -1, -1, 1, 1, -1) && \text{using } f(\lambda x) = f(x) \text{ for } \lambda > 0 \\ &= f(2, 0, 0, 2, 2, 0) && \text{using } f(x + e) = f(x) \\ &= f(1, 0, 0, 1, 1, 0), \end{aligned}$$

by using formulas from Section 1.3. As Figure 4 suggested, $f(y) = A$, since y is a profile only containing the three preferences (abc), (bca), (cab) exactly one time. We can use the symmetry of f to show $f(y) = A$. To see this, assume without loss of generality $f(y) = \{a\}$, use $\sigma = (a\ b\ c)$, because of symmetry we will get $f(M_\sigma(y)) = \sigma(f(y)) = \{b\}$. But notice $M_\sigma(y) = y$, so we also have $f(M_\sigma(y)) = f(y) = \{a\}$, contradiction. In other cases, we can still use $\sigma = (a\ b\ c)$ to deduce a contradiction.

If $\lambda = 0$, we have $f(y) = f(1, 1, 1, 1, 1, 1) = A$, this is trivial. If $\lambda < 0$, we have $f(y) = f(-1, 1, 1, -1, -1, 1) = f(0, 2, 2, 0, 0, 2) = f(0, 1, 1, 0, 0, 1)$. Similarly, we have $f(y) = A$. Thus

$$D(x) = 0 \quad \Rightarrow \quad f(x) = A.$$

□

3.2.2 Non-imposition

Now, we start the proof of the general dependency lemma. We will **prove the dependency lemma by contradiction**. Let's say (11) is false. Then assume $m \geq 2$ ($m = 1$ is trivial) and without loss of generality there is a $x^0 \in R$ with

$$f(x^0) = \{a_1, a_2, \ldots, a_r\} \quad \text{for some} \quad r < m. \tag{12}$$

With (12), we firstly show that f is non-imposing, which means that f could then choose any alternative uniquely with some profiles. Later we will then use the non-imposition to show Equation (16).

Let's define a profile

$$y^1 := x^0 + \sum_{j=2}^{r} M_{(jm)}(x^0) \tag{13}$$

where $M_{(jm)} \in \{0,1\}^{m! \times m!}$ denotes the permutation matrix which permutates only the alternative j and m. For $r = 1$, we define $y^1 = x^0$. We notice that $y^1 \in R$, (R defined in Equation (11)) since

$$\begin{aligned}
D(y^1) &= D(x^0 + \sum_{j=2}^{r} M_{(jm)}(x^0)) \\
&= D(x^0) + \sum_{j=2}^{r} D(M_{(jm)}(x^0)) && \text{by linearity} \\
&= D(x^0) + \sum_{j=2}^{r} M_{(jm)} D(x^0) \\
&= 0 && \text{since } D(x^0) = 0
\end{aligned}$$

In addition to that, we have $f(y^1) = \{a_1\}$, since y^1 is obtained by keep adding preferences so that each alternative will be excluded once except a_1. Formally, we have

$$\begin{aligned}
f(y^1) &= f(x^0 + \sum_{j=2}^{r} M_{(jm)}(x^0)) \\
&= f(x^0) \cap \left(\bigcap_{j=2}^{r} f(M_{(jm)}(x^0)) \right) && \text{by consistency} \\
&= A^r \cap \left(\bigcap_{j=2}^{r} ((A^r \setminus \{a_j\}) \cup \{a_m\}) \right) && \text{by symmetry. } A^r := \{a_1, \ldots, a_r\} \\
&= (A^* \setminus \{a_m\}) \cap \left(\bigcap_{j=2}^{r} (A^* \setminus \{a_j\}) \right) && \text{with } A^* := A^r \cup \{a_m\} \\
&= A^* \setminus \left(\{a_m\} \cup \left(\bigcup_{j=2}^{r} \{a_j\} \right) \right) = A^* \setminus (A^* \setminus \{a_1\}) \\
&= \{a_1\}.
\end{aligned}$$

With the same technique, we can also create profiles in R to choose other alternatives uniquely.

We now define the objects $R_i := \{x \in R : a_i \in f(x)\}$. As already shown in Section 1.4, we know that R_i is $\mathbb{Q}$-convex. Using lemma 1.9, we further conclude that $\overline{R}_i$ is convex. It is clear that

$$\bigcup_{i=1}^{m} \overline{R}_i = \overline{R}. \tag{14}$$

Next,

$$M_\sigma(\overline{R}_i) = \overline{R}_{\sigma(i)}, 1 \leq i \leq m \tag{15}$$

by symmetry. By lemma 1.10 we know that there exist some i so that $\mathbf{dim}\, \overline{R}_i = \mathbf{dim}\, \overline{R}$, thus $\mathbf{int}_{\overline{R}}\, \overline{R}_i \neq \varnothing$ for this i. Because of symmetry, we have

$$\mathbf{int}_{\overline{R}}\, \overline{R}_i \neq \varnothing \quad \text{for } 1 \leq i \leq m. \tag{16}$$

To get more intuition, we can refer to Equation (14), Equation (15), and Equation (16) with respect to Figure 4. Note that the visualizing is tricky, since we are proving by contradiction. Thus not everything can be shown, due to the fact that not everything is correct. Nevertheless, some equations are in fact true and can be visualized. Notice also in Figure 4 R is the whole domain of the function, not the same as the one defined in Equation (11), but this is not a problem since we can prove the equations in both cases.

3.2.3 No Intersection of the Preimage-cones

Next, we want to show that R_is can be represented by hyperplanes, using the separation theorem. In order to use the separation theorem, we need to firstly show that the interior of the R_is do not intersect.

We will prove this by another contradiction. Suppose that $\mathbf{int}_{\overline{R}}\,\overline{R}_1 \cap \mathbf{int}_{\overline{R}}\,\overline{R}_2 \neq \varnothing$. Then we choose $x^1 \in \mathbb{Q}^{m!} \cap \mathbf{int}_{\overline{R}}\,\overline{R}_1 \cap \mathbf{int}_{\overline{R}}\,\overline{R}_2$ and let $z(\lambda) := \lambda y^1 + (1-\lambda)x^1$ (with y^1 defined in Equation (13)) for each rational λ with $0 < \lambda < 1$. Then for all sufficient small such λ,

$$\begin{aligned} z(\lambda) &\in \mathbb{Q}^{m!} \cap \mathbf{int}_{\overline{R}}\,\overline{R}_1 \cap \mathbf{int}_{\overline{R}}\,\overline{R}_2 && z(\lambda) \text{ close enough to } x^1 \\ &= \mathbb{Q}^{m!} \cap \mathbf{int}_{\overline{R}}\,\overline{\mathbf{cvx}\ R_1} \cap \mathbf{int}_{\overline{R}}\,\overline{\mathbf{cvx}\ R_2} && \text{since } \overline{R}_i \text{ are convex} \\ &= \mathbb{Q}^{m!} \cap \mathbf{int}_{\overline{R}}\,\mathbf{cvx}\ R_1 \cap \mathbf{int}_{\overline{R}}\,\mathbf{cvx}\ R_2 && \text{by lemma 1.7} \\ &= \mathbf{int}_{\overline{R}}\,R_1 \cap \mathbf{int}_{\overline{R}}\,R_2 && \text{by lemma 1.8} \\ &\subseteq R_1 \cap R_2. \end{aligned}$$

Hence $\{a_1, a_2\} \in f(z(\lambda))$. But on the other hand, $f(z(\lambda)) = f(\lambda y^1 + (1-\lambda)x^1) = f(y^1) \cap f(x^1) = \{a_1\}$ by consistency. This is a contradiction. The symmetry of f implies

$$\mathbf{int}_{\overline{R}}\,\overline{R}_i \cap \mathbf{int}_{\overline{R}}\,\overline{R}_j = \varnothing \quad \text{for all } i \neq j. \tag{17}$$

The non-intersection of preimage-cones is visualized in Figure 4. Now we are ready to use the separation theorem.

3.2.4 Hyperplane-representation of the Preimage-cones

In this subsubsection, we aim at giving the hyperplane-representation of the preimage-cones (Equation (23)).

As already shown above, the interior (with respect to $\overline{R}$) of any two different convex sets R_i and R_j doesn't intersect, so the separation theorem for convex sets (see sec 2.5 of [BV04]) implies that for each $i \neq j$ there exists a normal vector $u^{ij} \in \overline{R}$ with $\left\|u^{ij}\right\| = 1$ such that

$$u^{ij} \cdot x \geq 0 \quad \text{for all } x \in \overline{R}_i \quad \text{and} \quad u^{ij} \cdot x \leq 0 \quad \text{for all } x \in \overline{R}_j. \tag{18}$$

Our goal next is to show the uniqueness of the hyperplanes, or equivalently, the uniqueness of u^{ij}. To reach the goal, we will firstly prove that $\overline{R}_i$ is restricted by hyperplanes.

For $i > j$, choose $u^{ij} = -u^{ji}$, so (18) holds for all $i \neq j$. Now we define $S_i := \{x \in \overline{R} : u^{ij} \cdot x \geq 0, \text{ for all } j \neq i\}$. Then

$$\varnothing \overset{(16)}{\neq} \mathbf{int}_{\overline{R}}\,\overline{R}_i \overset{(*)}{\subseteq} \{x \in \overline{R} : u^{ij} \cdot x > 0 \quad \text{for all } i \neq j\} = \mathbf{int}_{\overline{R}}\,S_i \tag{19}$$

and further

$$\mathbf{int}_{\overline{R}}\,S_i = \{x \in \overline{R} : u^{ij} \cdot x > 0 \quad \text{for all } i \neq j\} \subseteq \overline{R} \setminus \bigcup_{j \neq i} \overline{R}_j \subseteq \overline{R}_i. \tag{20}$$

(Notice that in $(*)$, we change from $\geq$ to $>$ because we are considering the interior.) Thus

$$\mathbf{int}_{\overline{R}}\,\overline{R}_i \subseteq \mathbf{int}_{\overline{R}}\,S_i \subseteq \overline{R}_i, \tag{21}$$

which also means

$$\overline{\mathbf{int}_{\overline{R}}\,\overline{R}_i} \subseteq \overline{\mathbf{int}_{\overline{R}}\,S_i} \subseteq \overline{R}_i. \tag{22}$$

We also notice that $\overline{\mathbf{int}_{\overline{R}}\,\overline{R}_i} = \overline{R}_i$ and $\overline{\mathbf{int}_{\overline{R}}\,S_i} = \overline{S}_i$ according to lemma 1.7. Together with (22) we get $\overline{R}_i \subseteq S_i \subseteq \overline{R}_i$, which means $\overline{R}_i = S_i$.

So we finally showed that $\overline{R}_i$ could be represented by hyperplanes, that is

$$\overline{R}_i = \{x \in \overline{R} : u^{ij} \cdot x \geq 0 \quad \text{for all } j \neq i\}. \tag{23}$$

Since $m \geq 2$, $\overline{R}_i$ has some face of dimension $\mathbf{dim}\ \overline{R} - 1$ by (23). Hence for some $j \neq i$, $\mathbf{dim}\ (\overline{R}_i \cap \overline{R}_j) = \mathbf{dim}\ \overline{R} - 1$. Again, by symmetry this holds for all $i \neq j$. Recall the fact that hyperplanes also have dimension $\mathbf{dim}\ \overline{R} - 1$, so the hyperplanes must be unique, which means u^{ij} is unique. It follows that

$$M_\sigma(u^{ij}) = u^{\sigma(i)\sigma(j)} \tag{24}$$

for all permutations σ.

Notice that Equation (23), (24), and the face dimension are illustrated in Figure 4 (This will be discussed in more detail later in Section 3.3).

3.2.5 Symmetry of Normal Vectors

The next step is to show that $u^{12} + u^{23} + u^{31} = 0$, that is, the symmetry of normal vectors.

We now assume $m \geq 3$ (for $m = 1, 2$, the dependency lemma is obvious). Let's choose $x^0 \in \mathbf{int}_{\overline{R}}\,\overline{R}_3$ and let $y := x^0 + M_{(12)}(x^0)$. Then

$$\begin{aligned} f(y) &= f(x^0 + M_{(12)}(x^0)) \\ &= f(x^0) \cap \big(\sigma_{(12)}(f(x^0))\big) && \text{by consistency and symmetry} \\ &= \{a_3\} \cap \{a_3\} = \{a_3\}. \end{aligned}$$

Thus $y \in \mathbf{int}_{\overline{R}}\,\overline{R}_3$. We also notice that

$$M_{(12)}(y) = M_{(12)}\left(x^0 + M_{(12)}(x^0)\right) = M_{(12)}(x^0) + M_{(12)}M_{(12)}(x^0) = M_{(12)}(x^0) + x^0 = y. \tag{25}$$

Thus

$$\begin{aligned} u^{12} \cdot y &\overset{(25)}{=} u^{12} \cdot \big(M_{(12)}(y)\big) && \text{notice "$\cdot$" is the dot product} \\ &= \big(M_{(12)}u^{12}\big) \cdot y && \text{symmetry of } M_{(12)} \\ &\overset{(24)}{=} u^{21} \cdot y \\ &= -u^{12} \cdot y && u^{ij} = -u^{ji} \quad \text{for } i > j \end{aligned}$$

which means $u^{12} \cdot y = 0$, thus y lies in the hyperplane which separates $\overline{R}_1$ and $\overline{R}_2$. So $y \in \mathbf{aff}\ (\overline{R}_1 \cap \overline{R}_2)$. Recall that $y \in \mathbf{int}_{\overline{R}}\,\overline{R}_3$, using (17), we deduce that $y \notin (\overline{R}_1 \cap \overline{R}_2)$. So we conclude that $y \in \big(\mathbf{aff}\ (\overline{R}_1 \cap \overline{R}_2)\big) \setminus (\overline{R}_1 \cap \overline{R}_2)$. The existence of such y ensures that $\overline{R}_1 \cap \overline{R}_2$ has a $(\mathbf{dim}\ \overline{R} - 2)$ dimensional face of form $\overline{R}_1 \cap \overline{R}_2 \cap \overline{R}_k$. By symmetry, we may assume $k = 3$. Now considering $u^{12}, u^{23}, u^{31} \in \overline{R}$. We know the three normal vectors are orthogonal to $\overline{R}_1 \cap \overline{R}_2 \cap \overline{R}_3$, so

$$u^{12}, u^{23}, u^{31} \in (\overline{R}_1 \cap \overline{R}_2 \cap \overline{R}_3)^{\perp}. \tag{26}$$

Notice

$$\mathbf{dim}\ (\overline{R}_1 \cap \overline{R}_2 \cap \overline{R}_3)^{\perp} = \mathbf{dim}\ \overline{R} - \mathbf{dim}\ (\overline{R}_1 \cap \overline{R}_2 \cap \overline{R}_3) = \mathbf{dim}\ \overline{R} - (\mathbf{dim}\ \overline{R} - 2) = 2. \tag{27}$$

Now (26) and (27) together suggest that u^{12}, u^{23}, u^{31} must be dependent since they are 3 vectors in a 2-dimensional space. So there exists $\lambda, \lambda', \lambda''$ not all zero with

$$\lambda u^{12} + \lambda' u^{23} + \lambda'' u^{31} = 0. \tag{28}$$

Next, we argue that u^{12}, u^{23}, u^{31} cannot all be equal. If they are all equal, then assume $i, j \notin \{1, 2, 3\}$, use $\sigma = (1\ i)$ we get $M_\sigma(u^{12}) = u^{i2}$ by (24). On the other hand, we have $M_\sigma(u^{12}) = M_\sigma(u^{23}) = u^{23}$. Thus $u^{i2} = u^{23} = u^{12} = u^{31}$. Then we use $\sigma' = (2\ j)$ and get $M_{\sigma'}(u^{i2}) = u^{ij}$. Notice again $u^{i2} = u^{31}$, we therefore have $M_{\sigma'}(u^{i2}) = M_{\sigma'}(u^{31}) = u^{31} = u^{12}$. So we conclude that $u^{ij} = u^{12}$ for all $i \neq j$. If this is true, then there could be at most two distinct regions $\overline{R}_i$, which contradicts the assumption $m \geq 3$. Therefore, we conclude that u^{12}, u^{23}, u^{31} are not all equal.

Let $\sigma = (1\ 2\ 3)$. We define $\tilde{u}^{ij} \in \mathbb{R}^3$ to be the restriction of $u^{ij} \in \mathbb{R}^{m!}$ with respect to $p \in \mathscr{P}$ which satisfies

$$\tilde{u}^{ij} = (\tilde{u}^{ij}_1, \tilde{u}^{ij}_2, \tilde{u}^{ij}_3)^T \quad \text{with } \tilde{u}^{ij}_1 = u^{ij}_p, \quad \tilde{u}^{ij}_2 = u^{ij}_{\sigma(p)}, \quad \tilde{u}^{ij}_3 = u^{ij}_{\sigma^2(p)}. \tag{29}$$

Here we use u^{ij}_p to denote the p-th entry of u^{ij}. Since u^{12}, u^{23}, u^{31} are not all equal, we can find such p so that $\tilde{u}^{12}, \tilde{u}^{23}, \tilde{u}^{31}$ are not all equal. It is clear that these restrictions, namely $\tilde{u}^{12}, \tilde{u}^{23}, \tilde{u}^{31}$, are also dependent, hence they lie in a 2-dimensional plane through the origin and they are permutated in a 3-cycle by the corresponding restriction of M_σ, which is just a rotation in $\mathbb{R}^3$ having axis $(1, 1, 1)^T$.

So we conclude $\tilde{u}^{12}, \tilde{u}^{23}, \tilde{u}^{31}$ are distinct and $\tilde{u}^{12} + \tilde{u}^{23} + \tilde{u}^{31} = 0$. Because of (28) we get

$$\lambda \tilde{u}^{12} + \lambda' \tilde{u}^{23} + \lambda'' \tilde{u}^{31} = 0. \tag{30}$$

Due to the fact that $\tilde{u}^{12}, \tilde{u}^{23}, \tilde{u}^{31}$ lie symmetrically in a 2-dim subspace, it follows that $\lambda = \lambda' = \lambda'' \neq 0$. (If not, Equation (30) can obviously not hold.) So

$$(28) \quad \Rightarrow \quad \lambda(u^{12} + u^{23} + u^{31}) = 0 \quad \Rightarrow \quad u^{12} + u^{23} + u^{31} = 0. \tag{31}$$

The symmetry of the normal vectors is illustrated in Section 3.3.

3.2.6 The Contradiction

In the next step, we would use (31) to deduce a contradiction against (12) (in (12) we assumed the existence of such $x^0 \in R$ so that $f(x^0) = \{a_1, a_2, \dots, a_r\}$ with some $r < m$), which would imply that the dependency lemma is true.

Consider M_σ where σ fixes 1 and 2. By (24), we have $M_\sigma(u^{12}) = u^{12}$. Hence, for any preference order p in which a_1 is ranked i-th and a_2 is ranked j-th, $u_p^{ij} \in \mathbb{R}$ has the same value. Let's define

$$\mathscr{P}_{ij} := \{p \in \mathscr{P} : a_1 \text{ is ranked i-th and } a_2 \text{ is ranked j-th}\} \tag{32}$$

and

$$s_{ij} := u_p^{12} \in \mathbb{R} \quad \text{for } p \in \mathscr{P}_{ij}, \text{ and let } s_{ii} = 0. \tag{33}$$

First, we will claim that

$$s_{ij} = -s_{ji} \tag{34}$$

since

$$M_{\sigma(12)} \cdot u^{12} = M_{\sigma(12)} \cdot \begin{pmatrix} \vdots \\ u_p^{12} = s_{ij} \\ \vdots \\ u_{\sigma(12)(p)}^{12} = s_{ji} \\ \vdots \end{pmatrix} = \begin{pmatrix} \vdots \\ u_{\sigma(12)(p)}^{12} = s_{ji} \\ \vdots \\ u_p^{12} = s_{ij} \\ \vdots \end{pmatrix}$$

$$= u^{21} = -u^{12} = \begin{pmatrix} \vdots \\ -u_p^{12} = -s_{ij} \\ \vdots \\ -u_{\sigma(12)(p)}^{12} = -s_{ji} \\ \vdots \end{pmatrix} \quad \text{for any } p \in \mathscr{P}_{ij}.$$

Further (31) implies

$$s_{ij} + s_{jk} + s_{ki} = 0 \tag{35}$$

for all distinct i, j, k, since

$$u^{12} = \begin{pmatrix} \vdots \\ u_p^{12} = s_{ij} \\ \vdots \\ u_{\sigma(123)(p)}^{12} = s_{jk} \\ \vdots \\ u_{\sigma^2_{(123)}(p)}^{12} = s_{ki} \\ \vdots \end{pmatrix} = \begin{pmatrix} \vdots \\ u_p^{12} = s_{ij} \\ \vdots \\ u_p^{23} = s_{jk} \\ \vdots \\ u_p^{31} = s_{ki} \\ \vdots \end{pmatrix} \quad \text{for any } p \in \mathscr{P}_{ij}.$$

Now consider any $x \in \overline{R} = \{x \in \mathbb{R}^{m!} : D(x) = 0\}$, we have

$$\begin{aligned}
u^{12} \cdot x &= \sum_{p \in \mathscr{P}} u^{12}_p x_p = \sum_{1 \le i,j \le m} \sum_{p \in \mathscr{P}_{ij}} u^{12}_p x_p = \sum_{1 \le i,j \le m} \sum_{p \in \mathscr{P}_{ij}} s_{ij} x_p \\
&= \sum_{1 \le i,j \le m} s_{ij} \sum_{p \in \mathscr{P}_{ij}} x_p \overset{(35)}{=} \sum_{1 \le i,j \le m} (-s_{jm} - s_{mi}) \sum_{p \in \mathscr{P}_{ij}} x_p \\
&\overset{(34)}{=} \sum_{1 \le i,j \le m} (s_{im} - s_{jm}) \sum_{p \in \mathscr{P}_{ij}} x_p \\
&= \sum_{1 \le i \le m} \sum_{1 \le j \le m} s_{im} \sum_{p \in \mathscr{P}_{ij}} x_p - \sum_{1 \le i \le m} \sum_{1 \le j \le m} s_{jm} \sum_{p \in \mathscr{P}_{ij}} x_p \\
&= \sum_{1 \le i \le m} s_{im} \underbrace{\left(\sum_{1 \le j \le m} \sum_{p \in \mathscr{P}_{ij}} x_p \right)}_{\substack{\text{how often } a_1 \text{ is ranked} \\ \text{in the i-th place}}} - \sum_{1 \le j \le m} s_{jm} \underbrace{\left(\sum_{1 \le i \le m} \sum_{p \in \mathscr{P}_{ij}} x_p \right)}_{\substack{\text{how often } a_2 \text{ is ranked} \\ \text{in the j-th place}}} \\
&= \sum_{1 \le i \le m} s_{im} D_{1i}(x) - \sum_{1 \le j \le m} s_{jm} D_{2j}(x) \\
&= 0 \qquad \text{since } D(x) = 0 \text{ for } x \in \overline{R}.
\end{aligned}$$

But $u^{12} \in \overline{R}$, which would imply $u^{12} \cdot u^{12} = 0$ by the calculation above. This is a contradiction since $u^{12} \cdot u^{12} > 0$ must hold as $u^{12} \neq 0$. Therefore (11) must be true, so we finally proved the dependency lemma (lemma 3.3).

3.2.7 A Central Result

The next step is to use the dependency lemma to obtain the central result that $f(x)$ only depends on $D(x)$.

Corollary 3.4. *For any symmetric and consistent SCF f and any profile x, $f(x)$ depends only on $D(x)$.*

Proof. Consider any two profiles x, x'. If $D(x) = D(x')$, then $D(x) - D(x') = D(x - x') = 0$ by linearity of D. Thus $x - x' \in R$, which means $f(x - x') = A$ by dependency lemma. So

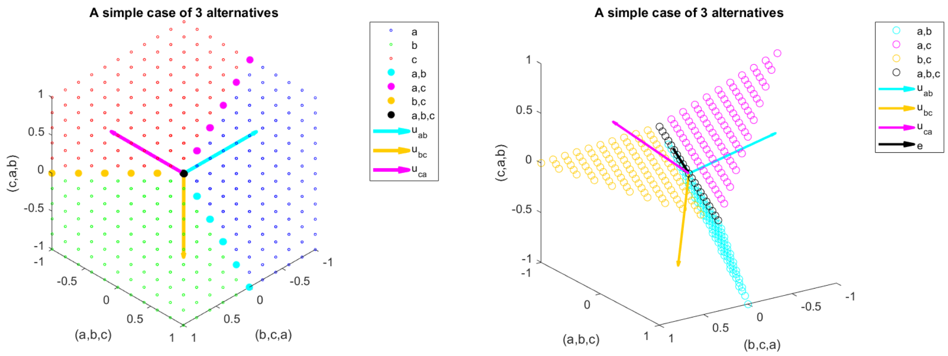
$$f(x) = f(x' + x - x') = f(x') \cap f(x - x') = f(x') \cap A = f(x'). \tag{36}$$

□

As a result, it showed that we only need to consider the image of D as the domain of such symmetric and consistent f, instead of $\mathbb{Q}^{m!}$.

3.3 Visualizing the Normal Vectors

Before continuing with the proof, we would like to present some illustrative examples by plotting the normal vectors for Borda's rule.



(a) view 1 (b) view 2

Figure 5: Normal vectors

Recalling the example from Section 2.2, where we have $A = \{a, b, c\}$ and we only consider three non-zero components (abc), (bca) and (cab). This time, we emphasize the hyperplanes and the normal vectors. Figure 5a is obtained from Figure 4a by adding the three normal vectors u^{ab}, u^{bc} and u^{ca}. Figure 5b is an adjusted view of Figure 5a, where we ignore the points selecting only one alternative and concentrate on the separating planes and normal vectors. In this example, we choose Borda's rule.

We now define $R_i := \{x \in \mathbb{Q}^{m!}, a_i \in f(x)\}$ and R to be the domain of f. $\overline{R}_i$ is the closure of R_i. As Figure 5a shows, we see $\overline{R}_a$, $\overline{R}_b$ and $\overline{R}_c$ are all convex cones, and they are nicely symmetric. Further we see $\overline{R}_a \cup \overline{R}_b \cup \overline{R}_c = \overline{R}$. Moreover, $\mathbf{int}\overline{R}_i \neq \varnothing$ and $\mathbf{int}\overline{R}_i \cap \mathbf{int}\overline{R}_j = \varnothing$ for $i, j \in \{a, b, c\}$, $i \neq j$. The hyperplanes, which separate $\overline{R}_i$ and $\overline{R}_j$, are presented by the normal vectors drawn in cyan, magenta, and brown. Notice they have one dimension less than $\overline{R}_i$, and they are unique. The points on the hyperplanes are presented in the same colors as their normal vectors'. The hyperplane, which separates R_i and R_j is made of points that select exactly these two alternatives i and j.

From Figure 5a, we can observe that, for example, $\overline{R}_a = \{x \in \overline{R} : u^{ab} \cdot x \geq 0 \text{ and } u^{ac} \cdot x \geq 0\}$. Notice that u^{ac} is not on the figure, but we know $u^{ac} = -u^{ca}$. This is Equation (23).

Next, we can check, for example, $M_{(abc)}u^{ab} = u^{bc}$, this is a special case of Equation (24). Notice, with Figure 5b, we can verify that the three normal vectors are permutated in a 3-cycle in a 2-dimensional plane through the origin, with rotation axis $(1, 1, 1)^T$. Further, we observe that $u^{ab} + u^{bc} + u^{ca} = 0$ since the three vectors are symmetric. This is what we try to prove in Section 3.2.5.

3.4 The Final Step

As discussed earlier, we will concentrate on the image of D, that is, by definition, all rational, linear combinations of $m \times m$ permutation matrices. But before we move on, we should shortly review where we stand now:

$$\begin{tikzcd}
x \in \mathbb{Q}^{m!} \arrow[r, "f"] \arrow[d, "D\ (\text{restriction})"'] & \mathcal{P}(A) \setminus \varnothing \\
D(x) \in \mathbb{Q}^{m \times m} \arrow[ur, bend right, "f^s"'] &
\end{tikzcd}$$

As the graph above suggested, the scoring rule f^s depends only on $D(x)$, by definition. The dependency lemma allows us now to move from $\mathbb{Q}^{m!}$ to the image of D, that is $\mathbb{Q}^{m \times m}$, which is a much smaller domain for big m. Notice for $x \in \mathbb{Q}^{m!}$, we have information on how many times each preference is chosen. With this information, we can then easily count how many times each alternative is ranked in each position, but we can not deduce the information the other way around. So what we achieved, with the help of the dependency lemma, is a more restricted domain, and this is also the domain for scoring rules. Our final goal would be to find the scoring vector s. But first, let's take a deeper look at the domain $D(x)$ by recalling the Birkhoff-von Neumann theorem:

Theorem 3.5 (Birkhoff-von Neumann theorem [Bir46; Neu53])**.** *A matrix M is doubly stochastic if and only if it is a convex combination of permutation matrices.*

Proof. See [Bir46; Neu53]. □

Using Birkhoff-von Neumann theorem, we can deduce the following corollary, which characterizes our new domain.

Corollary 3.6. *Linear combinations of permutation matrices are precisely these matrices with constant row and column sums.*

Proof. The linear combination of permutation matrices have constant row and column sums is obvious. For the other direction, consider any matrix M with constant row and column sums k. Then, using the Birkhoff-von Neumann theorem, we know there is a convex combination for $\frac{1}{k}M$ so that:

$$\sum_{i=1}^{l} c_i P_i = \frac{1}{k} M.$$

Where P_i are permutation matrices, since $\frac{1}{k}M$ is doubly stochastic. Clearly $\sum_{i=1}^{l} kc_i P_i$ is the linear combination for M. □

3.4.1 New Domain, Same Approach

By Corollary 3.6, we will, from now on, regard all rational $m \times m$ matrices with constant row and column sums, denoted by $\mathscr{D}$, as the effective domain of f.

Since the trivial function is a (simple) scoring function, we may assume that f is nontrivial. **Notice we can consider any $D \in \mathscr{D}$ as a vector in $\mathbb{Q}^{m^2}$ by simply expand it row by row.**

Notice that in the new matrix domain, **whenever** we write $D \cdot u$, although D and u are both matrices, we compute the **dot product** here, by expanding the both matrices row by row. It is **not** the matrix product! (The advantage of using " $\cdot$ " to denote the dot product of matrices is to keep the notation consistent here, and we want to emphasize that the approach is exactly the same as earlier.)

Therefore, we can apply the same argument as before to find separating vectors $u^{ij} \in \mathscr{D}$ satisfying $u^{ij} = -u^{ij}$ and relations (18), (23) and (24). It is technically the same as before so we only give a sketch here.

Firstly redefine $R := \mathscr{D}$, $R_i := \{D \in \mathscr{D} : a_i \in f(D)\}$. Assume for some $x^0 \in R$, we have $f(x^0) = \{a_1, \ldots, a_r\}$ with $r < m$. (Notice if such x^0 does not exist, it is a trivial scoring rule.) Then we can prove there are other profiles in R choosing a_i alone, for $i \in \{1, \ldots, m\}$. Thus we proved the existence of distinct $\mathbb{Q}$-convex sets R_i and further $\mathbf{int}_{\overline{R}} \overline{R}_i \neq \varnothing$, $\mathbf{int}_{\overline{R}} \overline{R}_i \cap \mathbf{int}_{\overline{R}} \overline{R}_j = \varnothing$. Then use the separation theorem to obtain the normal vector of the hyperplane, namely $u^{ij} \in \overline{R}$, which separates $\overline{R}_i$ and $\overline{R}_j$. Then show $\overline{R}_i$ is given by $\overline{R}_i = \{x \in \overline{R} : u^{ij} \cdot x \geq 0, \text{ for all } j \neq i\}$. Again we can prove the equality $M_\sigma(u^{ij}) = u^{\sigma(i)\sigma(j)}$.

However, we now interpret M_σ as a permutation matrix that interchanges the rows prescribed by σ. Since $M_{(ij)}(u^{ij}) = u^{ji} = -u^{ij}$, it follows

$$u^{ij} = \begin{pmatrix} 0 \\ \vdots \\ s^T \\ \vdots \\ -s^T \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} 0, \ldots, 0 \\ \vdots \\ s_1, \ldots, s_m \\ \vdots \\ -s_1, \ldots, -s_m \\ \vdots \\ 0, \ldots, 0 \end{pmatrix}, \tag{37}$$

that is, u^{ij} has in the i-th row $s = (s_1, \ldots, s_m)$ and in the j-th row $-s$, and all other rows are zero.

3.4.2 An Example of the new Domain using Borda's rule

In this subsubsection, we would like to give an illustrative example for the new matrix domain using Borda's rule.

Assume we have again three alternatives $A = \{a, b, c\}$. By Corollary 3.4, we consider the domain $\mathbb{Q}^{3\times 3}$. First, we analyze how the domain is divided according to its outcome. More specifically, fixing i, we want to find out for what kind of matrix $D \in \mathbb{Q}^{3\times 3}$, we have $D \in R_i$. For instance, let's find all matrices $D \in R_a$:

Consider any D with the following structure (Notice that the last column is not relevant for Borda, because it contributes zero score.):

$$D = \begin{pmatrix} d_1 & d_2 & \bullet \\ d_3 & d_4 & \bullet \\ d_5 & d_6 & \bullet \end{pmatrix}. \tag{38}$$

It is obvious that $D \in R_a$ if and only if

$$2 \cdot d_1 + d_2 > 2 \cdot d_3 + d_4 \quad \text{and} \quad 2 \cdot d_1 + d_2 > 2 \cdot d_5 + d_6.$$

Next, let's find out u^{ab} and u^{ac}. Notice in the new domain u^{ab} and u^{ac} are both element from $\mathbb{Q}^{3\times 3}$. Because of Equation (23) we know that $D \cdot u^{ab} > 0$ (this is the dot product!) for any $D \in R_a$. So we find (without normalizing and shifting)

$$u^{ab} = \begin{pmatrix} 2 & 1 & 0 \\ -2 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \tag{39}$$

since

$$\begin{aligned} & D \cdot u^{ab} = 2 \cdot d_1 + d_2 - 2 \cdot d_3 - d_4 > 0 \\ \Leftrightarrow \quad & 2 \cdot d_1 + d_2 > 2 \cdot d_3 + d_4. \end{aligned}$$

Similarly, we find (without normalizing and shifting)

$$u^{ac} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 0 & 0 \\ -2 & -1 & 0 \end{pmatrix}. \tag{40}$$

The key observation is that u^{ij} $(i > j)$ seems to have the following structure:

$$u^{ij} = \begin{pmatrix} \vdots \\ s^T \\ \vdots \\ -s^T \\ \vdots \end{pmatrix}, \tag{41}$$

with the scoring vector s on the i-th row and $-s$ on the j-th row. Other rows are zero. Recalling that our goal is to find the scoring vector, giving some convex and consistency structure. This observation suggests us to find the scoring vector from the normal vector u^{ij}.

3.4.3 Proof of Main Theorem (i)

Now we will complete the formal proof. We will now denote the dot product of two matrices A and B by $\langle A, B \rangle$ (to avoid misunderstanding), by simply expanding them row by row into vectors and computing the dot product. It is easy to see that $\left\langle u^{ij}, D\right\rangle \geq 0$ if and only if $D_i \cdot s \geq D_j \cdot s$, since

$$\left\langle u^{ij}, D\right\rangle = D_i \cdot s - D_j \cdot s.$$

Hence, by (23), $a_i \in f(D)$ implies $u^{ij} \cdot D = \left\langle u^{ij}, D\right\rangle \geq 0$ for all $j \neq i$, which is equivalent to $D_i \cdot s \geq D_j \cdot s$ for all $j \neq i$, as just showed above. This means $f \precsim f^s$.

If $f = f^s$, then we are done. Otherwise, suppose inductively that $f \prec f^{s^\alpha} \circ \cdots \circ f^{s^1} := g$ for real m-vectors $s^1, \ldots, s^\alpha$. Without loss of generality, let $D^0 \in \mathscr{D}$ such that

$$f(D^0) = \{a_1, a_2, \ldots, a_k\} \subset \{a_1, a_2, \ldots, a_l\} = g(D^0), \quad k < l \leq m. \tag{42}$$

Let $D^1 := D^0 + \sum_{j=2}^{k} M_{(jl)}(D^0)$ for $k \geq 2$ and let $D^1 = D^0$ if $k = 1$. Then a_1 is the unique winner under $f(D^1)$ by applying the same argument as in Section 3.2.2. However, $g(D^1)$ would choose every alternative in $\{a_1, \ldots, a_l\}$, since

$$\begin{aligned}
g(D^1) &= g(D^0 + \sum_{j=2}^{k} M_{(jl)}(D^0)) \\
&= g(D^0) \cap g(M_{(2l)}(D^0)) \cap g(M_{(3l)}(D^0)) \cap \cdots \cap g(M_{(kl)}(D^0)) \\
&= \{a_1, a_2, \cdots, a_l\} \cap \{a_1, a_2, \cdots, a_l\} \cap \cdots \cap \{a_1, a_2, \cdots, a_l\} \\
&= \{a_1, a_2, \cdots, a_l\}.
\end{aligned}$$

Now choose any D^2 such that $g(D^2) = \{a_2\}$. Then for all $n > 0$, $g(D^2 + nD^1) = \{a_2\}$ by consistency. Since $f \prec g$, $f(D^2 + nD^1) = \{a_2\}$, for all $n \geq 0$, while $f(D^1) = \{a_1\}$. Hence, this f is not continuous.

Therefore, we showed that

$$f \neq f^s \quad \Rightarrow \quad \text{f not continuous.} \tag{43}$$

The contraposition of (43) implies

$$\text{f continuous} \quad \Rightarrow \quad f = f^s. \tag{44}$$

Notice $f = f^s$ means f is a (simple) scoring rule. So we finally proved main theorem (i).

3.4.4 Proof of Main Theorem (ii)

Now define

$$R' := \{D \in \mathscr{D} : D_1 \cdot s^1 = D_2 \cdot s^1 > D_j \cdot s^1 \text{ for } j \geq 3, \text{ and } D_1 \cdot s^i = D_2 \cdot s^i \text{ for } i \geq 2\}. \tag{45}$$

We can image R' as a part of the domain in that a_1 and a_2 win in the first round and tie until the end. Further, we define $R'_i := \{D \in R' : a_i \in f(D)\}$, where $i = 1, 2$. For all $D \in R'$,

$f(D) \subseteq g(D) = \{a_1, a_2\}$ (g was defined earlier). Notice $R' = R'_1 \cup R'_2$ and $\overline{R'} = \overline{R'}_1 \cup \overline{R'}_2$. Further,

$$\begin{aligned}\overline{R'} &= \{D \in \overline{\mathscr{D}} : D_1 \cdot s^1 = D_2 \cdot s^1 \geq D_j \cdot s^1 \text{ for } j \geq 3, \text{ and } D_1 \cdot s^i = D_2 \cdot s^i \text{ for } i \geq 2\} \\ &\overset{(*)}{\supseteq} \{D \in \overline{\mathscr{D}} : \{a_1, a_2\} \subseteq g(D)\}.\end{aligned}$$

To see $(*)$, consider any $D \in \overline{\mathscr{D}}$, if a_1 and a_2 will be chosen by g, then they must win the first round, with potentially other winners sharing the first place with them. And in the rest of the voting process, they must get the same score to both survive. So these D are in $\overline{R'}$.

Then we can use the same approach as in Section 3.2 to show

$$\underbrace{\mathbf{int}_{\mathbf{aff}\ \overline{R'}}\ \overline{R'}_1}_{\neq\varnothing} \cap \underbrace{\mathbf{int}_{\mathbf{aff}\ \overline{R'}}\ \overline{R'}_2}_{\neq\varnothing} = \varnothing, \tag{46}$$

and $M_{(12)}(\overline{R'}_1) = \overline{R'}_2$. Hence, there is an $u \in \overline{R'}$ of form

$$u = \begin{pmatrix} t_1 & t_2 & \cdots & t_m \\ -t_1 & -t_2 & \cdots & -t_m \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix} \in \mathbb{R}^{m \times m} \tag{47}$$

such that $\overline{R'}_1 = \{D \in \overline{R'} : D_1 \cdot t \geq D_2 \cdot t\}$.

For any D, $a_1 \in f(D)$ implies $a_1 \in g(D)$. If a_1 is not the only alternative selected by $g(D)$, then without loss of generality $\{a_1, a_2\} \subseteq g(D)$, then $D \in \overline{R'}_1$ and $D_1 \cdot t \geq D_2 \cdot t$. Similarly, we have $D_1 \cdot t \geq D_j \cdot t$ for all j such that $a_j \in g(D)$. Hence $a_1 \in f^t \circ g(D)$. By symmetry, it follows that

$$f \precsim f^t \circ g = f^t \circ f^{s^\alpha} \circ f^{s^{\alpha-1}} \circ \cdots \circ f^{s^1}. \tag{48}$$

Now we only need to show that the composition series will terminate, namely, at some point, we have

$$f = f^{\tilde{t}} \circ \tilde{g}. \tag{49}$$

Moreover, $u \in \overline{R'}$ implies $t \cdot s^\beta = 0$ for $1 \leq \beta \leq \alpha$. To see this, consider any $D \in \overline{R'}$. By definition, we get $D_1 \cdot s^i = D_2 \cdot s^i$, for $i \in \mathbb{N}_{>0}$. In this case, we have $D_1 = t$ and $D_2 = -t$, thus $t \cdot s^i = -t \cdot s^i$. This shows the claim that $t \cdot s^\beta = 0$, which means that the scoring vectors stand orthogonal to each other.

Notice we may always choose t so that $|t| := \sum_{i=1}^m t_i = 0$. This can be done by simply shifting the vector t. This step is required because we only allow matrices with constant row and column sums. In this case, the row sum must be zero.

Therefore the construction must terminate in at most $m - 1$ steps with the conclusion that

$$f(D) = f^{s^\gamma} \circ f^{s^{\gamma-1}} \circ \cdots \circ f^{s^1}(D), \tag{50}$$

for some $s^1, \ldots, s^\gamma \in \mathbb{R}^m$ such that $\left|s^\beta\right| = 0$, and $s^\alpha \cdot s^\beta = 0$ for $1 \leq \alpha \neq \beta \leq \gamma$ and for all $D \in \mathscr{D}$. Notice that $s^\alpha \cdot s^\beta = 0$ means s^α and s^β stand orthogonal to each other. To see the $m-1$ step termination, notice that the scoring vector s^i has dimension m. And there could be at most m such orthogonal vectors in an m dimensional space. So we are finished.

3.5 Visualizing the Composed Rule

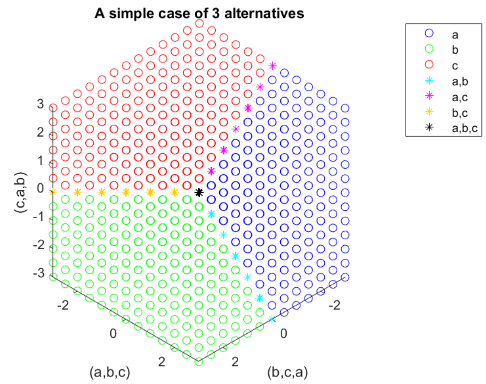
In this section, we want to visualize the composed scoring rule. The setting is almost the same as in Section 2.2, where we have $A = \{a, b, c\}$ and we only consider three preferences (abc), (bca), (cab). But now we choose f to be a composed scoring rule, namely, we first use Borda and then use plurality to break ties. Formally, we have $f = f^{(1,0,0)} \circ f^{(2,1,0)}$. The results are presented in Figure 6.

In Figure 6a, we plot Borda again for comparison, and then we plot plurality in Figure 6b. We see that the two simple scoring rules are "almost the same", in the sense that their images are the same if we ignore the rotation. Next, in Figure 6c, we add plurality on Borda. This figure could be interpreted as the following: The color of the outside circles represents the first-round result using Borda's rule. The inside circles represent the second-round result using plurality. However, according to the definition of composed scoring rules, if a unique winner exists, then it would be chosen till the end. So we only need to consider the points, which have ties in the first round. This leads to the simplified version, that is Figure 6d. Notice the plurality does break ties on the hyperplanes, except on the line λe. For example, the cyan points, which choose a and b by Borda in the first round, choose a ultimately by plurality. Similarly, the ultimate winner between b and c is b, between c and a is c. Notice the points on λe always choose $\{a, b, c\}$ using all composed scoring rules. The final result of the composed scoring rule is Figure 6e.

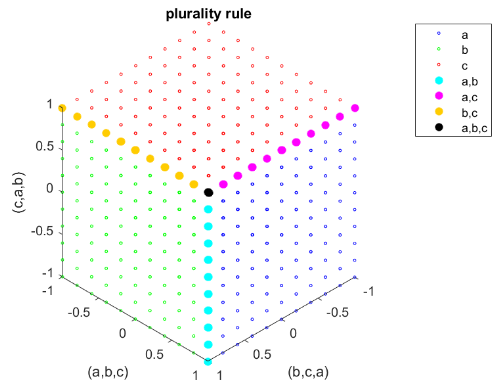
Notice in the final result there is no point choosing exactly two alternatives. So the boundary is "sharp". Such a structure is not achievable using only simple scoring rules. But, such construction is obviously symmetric and consistent. This would also suggest that only symmetry and consistency are not enough for characterizing simple scoring rules —– we still need continuity. Otherwise, it is the more general case of scoring rules — the composed scoring rules.

We also proved at the end that the construction $f = f^{s^\alpha} \circ \cdots \circ f^{s^1}$ must terminate in at most $m-1$ steps. In this example, we have $m = 3$, and we have a composition series of length 2.

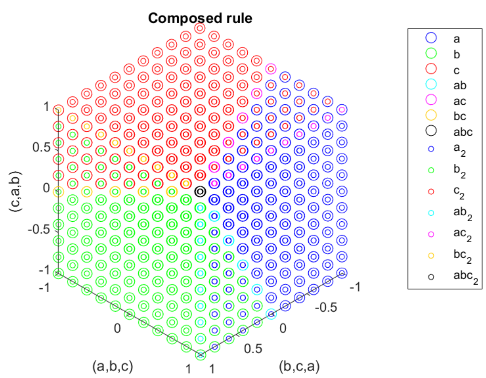
Notice that we have in this case $s_1 = (2,1,0)^T$ and $s_2 = (1,0,0)^T$. It holds that $s_1 \cdot s_2 \neq 0$ which seems at first to violate the last part of the proof, that the scoring vectors need to stand orthogonal to each other. But recall that the scoring vectors can be shifted without changing the outcome. Here we can therefore use $\tilde{s_1} = (2,1,0)^T - 2e = (0,-1,-2)^T$ and $\tilde{s_2} = s_2$ to get $\tilde{s_1} \cdot \tilde{s_2} = 0$.



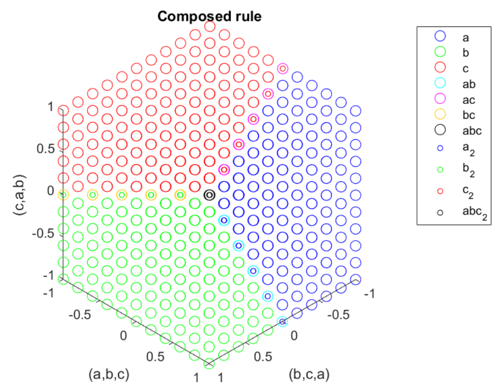
(a) Borda



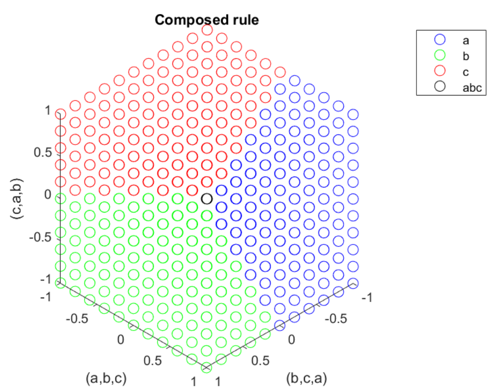
(b) Plurality



(c) Composed: Borda then Plurality



(d) Composed: Borda then Plurality, simplified



(e) Composed: Borda then Plurality, final

Figure 6: Visualization of the composed rule (Section 3.5)

3.6 Review of the Proof's Structure

Since the proof is quite complex, we would like to give a review of the whole proof to clarify its structure. Recall the two statements of the main theorem 3.1:

(i.) A SCF is symmetric, consistent, and continuous if and only if it is a (simple) scoring rule.

(ii.) A SCF is symmetric and consistent if and only if it is a composed scoring rule.

Again, the "if" direction is clear. To prove the "only if" direction, the first goal is to find the vector $s \in \mathbb{Q}^m$, satisfying

$$D_i(x) \cdot s \geq D_j(x) \cdot s \quad \forall a_i \in f(x),\ j \neq i,$$

for any symmetric and consistent SCF $f : \mathbb{Q}^{m!} \to \mathcal{P}(A) \setminus \varnothing$ and any profile $x \in \mathbb{Q}^m$ with a given alternative set $A = \{a_1, \ldots, a_m\}$ (for the function D, see Definition 1.4).

We found the vector s by investigating the preimage's structure of such symmetric and consistent SCFs. Given any such SCF f, to analyze its preimage, define $R_i = \{x \in \mathbb{Q}^{m!} : a_i \in f(x)\}$. Using consistency and symmetry, we showed that R_is are $\mathbb{Q}$-convex-cones, which lie symmetrically around the origin (Figure 4).

In order to represent the preimages-cones using the hyperplanes, we moved from $\mathbb{Q}$ to $\mathbb{R}$ (hyperplane separation theorems work only in $\mathbb{R}$), and considered $\overline{R}_i$s. We showed some properties of $\overline{R}_i$s such as:

- $\overline{R}_i$ are convex ($\mathbb{R}$-convex) cones
- $\bigcup_{i=1}^{m} \overline{R_i} = \overline{R}$
- $M_\sigma(\overline{R}_i) = \overline{R}_{\sigma(i)}$
- $\mathbf{int}_{\overline{R}}\, \overline{R}_i \neq \varnothing$
- $\mathbf{int}_{\overline{R}}\, \overline{R}_i \cap \mathbf{int}_{\overline{R}}\, \overline{R}_j = \varnothing, \quad i \neq j.$

Since the last property above holds, we then used the separation theorem for convex sets and proved that

- $\forall\, i \neq j \quad \exists!\, u^{ij} \in \overline{R},\ \left\|u^{ij}\right\| = 1 : \quad (u^{ij} \cdot x \geq 0\ \forall x \in \overline{R}_i) \ \wedge\ (u^{ij} \cdot x \leq 0\ \forall x \in \overline{R}_j)$
- $\overline{R}_i = \{x \in \overline{R} : u^{ij} \cdot x \geq 0 \quad \text{for all } j \neq i\}$
- $M_\sigma(u^{ij}) = u^{\sigma(i)\sigma(j)}$.

With the last property, by contradiction, we proved the dependency lemma, which states that $f(x)$ only depends on $D(x)$. Using the dependency lemma, we moved from $\mathbb{Q}^{m!}$ to $\mathbb{Q}^{m \times m}$.

Again in the new domain, all the properties above hold. The advantage of the new domain is that we can easily extract the scoring vector $s \in \mathbb{Q}^m$ from $u^{ij} \in \mathbb{Q}^{m \times m}$. We proved that

$$u^{ij} = \begin{pmatrix} \vdots \\ s^T \\ \vdots \\ -s^T \\ \vdots \end{pmatrix},$$

with s^T on the i-th row and $-s^T$ on the j-th row, all other rows are zeros. Denote $D \in \mathbb{Q}^{m \times m}$ as any element of our new domain, we have found such s which satisfies

$$D_i \cdot s \geq D_j \cdot s \quad \forall a_i \in f(D),\ j \neq i.$$

So we showed $f \precsim f^s$.

Next we showed that

$$f \neq f^s \quad \Rightarrow \quad f \text{ not continuous.}$$

Using its contraposition, we got

$$f \text{ continuous} \quad \Rightarrow \quad f = f^s.$$

So we finished the proof of (i.). To prove the second statement, suppose that

$$f \prec f^{s^\alpha} \circ \cdots \circ f^{s^1} := g.$$

Then we constructed a scoring vector t such that

$$f \precsim f^t \circ g.$$

The last step is to show that such construction must terminate in finite steps, that is

$$f = f^{\tilde{t}} \circ \cdots \circ g.$$

To see this, we proved that we can always construct scoring vectors which stand orthogonal to each other. It follows that the construction terminates in at most $m - 1$ steps, since there are at most m such orthogonal scoring vectors. So we are done.

Appendices

Matlab Source Code

This is the source code to generate Figure 4.

```
score_x=[2,1,0]; % score for the vector (1,0,0), representing the profile (abc)
score_y=[0,2,1]; % score for the vector (0,1,0), representing the profile (bca)
score_z=[1,0,2]; % score for the vector (0,0,1), representing the profile (cab)

for x=-3:0.5:3
    for y=-3:0.5:3
        for z=-3:0.5:3
            S=x*score_x+y*score_y+z*score_z; % calculating the score on point (x,y,z)
            if S(1)>S(2) && S(1)>S(3) % a get the highest score
                plot_a=plot3(x,y,z,'ob');
                hold on
            elseif S(2)>S(1) && S(2)>S(3) % b get the highest score
                plot_b=plot3(x,y,z,'og');
                hold on
            elseif S(3)>S(1) && S(3)>S(2) % c get the highest score
                plot_c=plot3(x,y,z,'or');
                hold on
            elseif S(1)==S(2) && S(1)>S(3) % a and b get the highest score
                plot_ab=plot3(x,y,z,'*c');
                hold on
            elseif S(1)==S(3) && S(1)>S(2) % a and c get the highest score
                plot_ac=plot3(x,y,z,'*m');
                hold on
            elseif S(2)==S(3) && S(2)>S(1) % b and c get the highest score
                plot_bc=plot3(x,y,z,'*','Color',[1 0.8 0]);
                hold on
```

```
            else % a, b and c get the highest score
                plot_abc=plot3(x,y,z,'*k');
                hold on
            end
        end
    end
end

axis equal
legend([plot_a,plot_b,plot_c,plot_ab,plot_ac,plot_bc,plot_abc], ...
    {'a','b','c','a,b','a,c','b,c','a,b,c'})
xlabel('(a,b,c)')
ylabel('(b,c,a)')
zlabel('(c,a,b)')
title('A simple case of 3 alternatives')
hold off
```

Source code to generate Figure 5a.

```
score_x=[2,1,0];
score_y=[0,2,1];
score_z=[1,0,2];

for x=-1:0.2:1
    for y=-1:0.2:1
        for z=-1:0.2:1
            S=x*score_x+y*score_y+z*score_z;
            if abs(S(1)-S(2))<0.05 && S(1)>S(3)
                plot_ab=plot3(x,y,z,'*c',LineWidth=2);
                hold on
            elseif abs(S(1)-S(3))<0.05 && S(1)>S(2)
                plot_ac=plot3(x,y,z,'*m',LineWidth=2);
                hold on
            elseif abs(S(2)-S(3))<0.05 && S(2)>S(1)
                plot_bc=plot3(x,y,z,'*','color',[1 0.8 0],LineWidth=2);
                hold on
            elseif S(1)>S(2) && S(1)>S(3)
                plot_a=plot3(x,y,z,'ob',LineWidth=0.5,MarkerSize=2);
```

```
                hold on
            elseif S(2)>S(1) && S(2)>S(3)
                plot_b=plot3(x,y,z,'og',LineWidth=0.5,MarkerSize=2);
                hold on
            elseif S(3)>S(1) && S(3)>S(2)
                plot_c=plot3(x,y,z,'or',LineWidth=0.5,MarkerSize=2);
                hold on
            else
                plot_abc=plot3(x,y,z,'*k',LineWidth=2);
                hold on
            end
        end
    end
end

% plot the normal vectors
u_ca=quiver3(0,0,0,-2/norm([-2,1,1]),1/norm([-2,1,1]),1/norm([-2,1,1]),'m','LineWidth',3);
u_ab=quiver3(0,0,0,1/norm([1,-2,1]),-2/norm([1,-2,1]),1/norm([1,-2,1]),'c','LineWidth',3);
u_bc=quiver3(0,0,0,1/norm([1,1,-2]),1/norm([1,1,-2]),-2/norm([1,1,-2]),'color',[1 0.8 0],'LineWidth',3);

axis equal
legend([plot_a,plot_b,plot_c,plot_ab,plot_ac,plot_bc,plot_abc, ...
    u_ab,u_bc,u_ca], ...
    {'a','b','c','a,b','a,c','b,c','a,b,c', ...
    'u_{ab}','u_{bc}','u_{ca}'})
xlabel('(a,b,c)')
ylabel('(b,c,a)')
zlabel('(c,a,b)')
title('A simple case of 3 alternatives')
hold off
```

Source code to generate Figure 5b.

```
score_x=[2,1,0];
score_y=[0,2,1];
score_z=[1,0,2];
```

```

for x=-1:0.1:1
    for y=-1:0.1:1
        for z=-1:0.1:1
            S=x*score_x+y*score_y+z*score_z;
            if abs(S(1)-S(2))<0.05 && S(1)>S(3)
                plot_ab=plot3(x,y,z,'oc');
                hold on
            elseif abs(S(1)-S(3))<0.05 && S(1)>S(2)
                plot_ac=plot3(x,y,z,'om');
                hold on
            elseif abs(S(2)-S(3))<0.05 && S(2)>S(1)
                plot_bc=plot3(x,y,z,'o','Color',[1 0.8 0]);
                hold on
            elseif abs(S(1)-S(2))<0.05 && abs(S(1)-S(3))<0.05
                plot_abc=plot3(x,y,z,'ok');
                hold on
            end
        end
    end
end

e=quiver3(0,0,0,1/norm([1,1,1]),1/norm([1,1,1]),1/norm([1,1,1]),'
    k','LineWidth',2);
u_ca=quiver3(0,0,0,-0.8165,0.4082,0.4082,'m','LineWidth',2);
u_ab=quiver3(0,0,0,0.4082,-0.8165,0.4082,'c','LineWidth',2);
u_bc=quiver3(0,0,0,0.4082,0.4082,-0.8165,'color',[1 0.8 0],'
    LineWidth',2);

axis equal
legend([plot_ab,plot_ac,plot_bc,plot_abc,u_ab,u_bc,u_ca,e], ...
    {'a,b','a,c','b,c','a,b,c','u_{ab}','u_{bc}','u_{ca}','e'})
xlabel('(a,b,c)')
ylabel('(b,c,a)')
zlabel('(c,a,b)')
title('A simple case of 3 alternatives')
hold off
```

Source code to generate plurality, Figure 6b.

```
for x=-1:0.2:1
```

```
    for y=-1:0.2:1
        for z=-1:0.2:1
            if abs(x-y)<0.02 && x>z
                plot_ab_2=plot3(x,y,z,'*c',LineWidth=2);
                hold on
            elseif abs(x-z)<0.02 && x>y
                plot_ac_2=plot3(x,y,z,'*m',LineWidth=2);
                hold on
            elseif abs(y-z)<0.02 && y>x
                plot_bc_2=plot3(x,y,z,'*','color',[1 0.8 0],
                   LineWidth=2);
                hold on
            elseif x>y && x>z
                plot_a_2=plot3(x,y,z,'ob',LineWidth=0.5,
                   MarkerSize=2);
                hold on
            elseif y>x && y>z
                plot_b_2=plot3(x,y,z,'og',LineWidth=0.5,
                   MarkerSize=2);
                hold on
            elseif z>x && z>y
                plot_c_2=plot3(x,y,z,'or',LineWidth=0.5,
                   MarkerSize=2);
                hold on
            else
                plot_abc_2=plot3(x,y,z,'*k',LineWidth=2);
                hold on
            end
        end
    end
end

axis equal
legend([plot_a_2,plot_b_2,plot_c_2,plot_ab_2,plot_ac_2,plot_bc_2,
   plot_abc_2], ...
    {'a','b','c','a,b','a,c','b,c','a,b,c'})
xlabel('(a,b,c)')
ylabel('(b,c,a)')
zlabel('(c,a,b)')
title('plurality rule')
```

```
hold off
```

Source code to generate composed rule, Figure 6c.

```
score_x=[2,1,0];
score_y=[0,2,1];
score_z=[1,0,2];

% plot Borda
for x=-1:0.2:1
    for y=-1:0.2:1
        for z=-1:0.2:1
            S=x*score_x+y*score_y+z*score_z;
            if abs(S(1)-S(2))<0.05 && S(1)>S(3)
                plot_ab=plot3(x,y,z,'oc',MarkerSize=8);
                hold on
            elseif abs(S(1)-S(3))<0.05 && S(1)>S(2)
                plot_ac=plot3(x,y,z,'om',MarkerSize=8);
                hold on
            elseif abs(S(2)-S(3))<0.05 && S(2)>S(1)
                plot_bc=plot3(x,y,z,'o','color',[1 0.8 0],
                    MarkerSize=8);
                hold on
            elseif S(1)>S(2) && S(1)>S(3)
                plot_a=plot3(x,y,z,'ob',MarkerSize=8);
                hold on
            elseif S(2)>S(1) && S(2)>S(3)
                plot_b=plot3(x,y,z,'og',MarkerSize=8);
                hold on
            elseif S(3)>S(1) && S(3)>S(2)
                plot_c=plot3(x,y,z,'or',MarkerSize=8);
                hold on
            elseif abs(S(1)-S(2))<0.02 && abs(S(1)-S(3))<0.02
                plot_abc=plot3(x,y,z,'ok',MarkerSize=8);
                hold on
            end
        end
    end
end

% plot plurality
```

```
for x=-1:0.2:1
    for y=-1:0.2:1
        for z=-1:0.2:1
            if abs(x-y)<0.02 && x>z
                plot_ab_2=plot3(x,y,z,'oc',MarkerSize=4);
                hold on
            elseif abs(x-z)<0.02 && x>y
                plot_ac_2=plot3(x,y,z,'om',MarkerSize=4);
                hold on
            elseif abs(y-z)<0.02 && y>x
                plot_bc_2=plot3(x,y,z,'o','color',[1 0.8 0],MarkerSize=4);
                hold on
            elseif x>y && x>z
                plot_a_2=plot3(x,y,z,'ob',MarkerSize=4);
                hold on
            elseif y>x && y>z
                plot_b_2=plot3(x,y,z,'og',MarkerSize=4);
                hold on
            elseif z>x && z>y
                plot_c_2=plot3(x,y,z,'or',MarkerSize=4);
                hold on
            else
                plot_abc_2=plot3(x,y,z,'ok',MarkerSize=4);
            end
        end
    end
end

axis equal
legend([plot_a,plot_b,plot_c,plot_ab,plot_ac,plot_bc,plot_abc, ...
    plot_a_2,plot_b_2,plot_c_2,plot_ab_2,plot_ac_2,plot_bc_2,plot_abc_2], ...
    {'a','b','c','ab','ac','bc','abc', ...
    'a_2','b_2','c_2','ab_2','ac_2','bc_2','abc_2'})
xlabel('(a,b,c)')
ylabel('(b,c,a)')
zlabel('(c,a,b)')
title('Composed rule')
hold off
```

Source code to generate composed rule, simplified, Figure 6d.

```
score_x=[2,1,0];
score_y=[0,2,1];
score_z=[1,0,2];

for x=-1:0.2:1
    for y=-1:0.2:1
        for z=-1:0.2:1
            S=x*score_x+y*score_y+z*score_z;
            if abs(S(1)-S(2))<0.05 && S(1)>S(3) % if a and b are tied
                plot_ab=plot3(x,y,z,'oc',MarkerSize=8);
                hold on
                if x>y % plot plurality
                    plot_a_2=plot3(x,y,z,'ob',MarkerSize=4);
                    hold on
                elseif y>x
                    plot_b_2=plot3(x,y,z,'og',MarkerSize=4);
                    hold on
                end
            elseif abs(S(1)-S(3))<0.05 && S(1)>S(2) % if a and c are tied
                plot_ac=plot3(x,y,z,'om',MarkerSize=8);
                hold on
                if x>z % plot plurality
                    plot_a_2=plot3(x,y,z,'og',MarkerSize=4);
                    hold on
                elseif z>y
                    plot_c_2=plot3(x,y,z,'or',MarkerSize=4);
                    hold on
                end
            elseif abs(S(2)-S(3))<0.05 && S(2)>S(1) % if b and c are tied
                plot_bc=plot3(x,y,z,'o','color',[1 0.8 0],MarkerSize=8);
                hold on
                if y>z % plot plurality
                    plot_b_2=plot3(x,y,z,'og',MarkerSize=4);
```

```
                        hold on
                    elseif z>y
                        plot_c_2=plot3(x,y,z,'or',MarkerSize=4);
                        hold on
                    end
                elseif S(1)>S(2) && S(1)>S(3)
                    plot_a=plot3(x,y,z,'ob',MarkerSize=8);
                    hold on
                elseif S(2)>S(1) && S(2)>S(3)
                    plot_b=plot3(x,y,z,'og',MarkerSize=8);
                    hold on
                elseif S(3)>S(1) && S(3)>S(2)
                    plot_c=plot3(x,y,z,'or',MarkerSize=8);
                    hold on
                elseif abs(S(1)-S(2))<0.02 && abs(S(1)-S(3))<0.02 %
                    if a,b and c are tied
                    plot_abc=plot3(x,y,z,'ok',MarkerSize=8);
                    hold on
                    if abs(x-y)<0.02 && abs(y-z)<0.02 % plot
                        plurality
                        plot_abc_2=plot3(x,y,z,'ok',MarkerSize=4);
                        hold on
                    end
                end
            end
        end
end

axis equal
legend([plot_a,plot_b,plot_c,plot_ab,plot_ac,plot_bc,plot_abc, ...
    plot_a_2,plot_b_2,plot_c_2,plot_abc_2], ...
    {'a','b','c','ab','ac','bc','abc', ...
    'a_2','b_2','c_2','abc_2'})
xlabel('(a,b,c)')
ylabel('(b,c,a)')
zlabel('(c,a,b)')
title('Composed rule')
hold off
```

Source code to generate composed rule, final, Figure 6e.

```
score_x=[2,1,0];
score_y=[0,2,1];
score_z=[1,0,2];

for x=-1:0.2:1
    for y=-1:0.2:1
        for z=-1:0.2:1
            S=x*score_x+y*score_y+z*score_z;
            if abs(S(1)-S(2))<0.05 && S(1)>S(3) % if a and b are tied
                if x>y % then use plurality to break tie
                    plot_a_2=plot3(x,y,z,'ob',MarkerSize=8);
                    hold on
                else
                    plot_b_2=plot3(x,y,z,'og',MarkerSize=8);
                    hold on
                end
            elseif abs(S(1)-S(3))<0.05 && S(1)>S(2) % if a and c are tied
                if x>z % then use plurality to break tie
                    plot_a_2=plot3(x,y,z,'ob',MarkerSize=8);
                    hold on
                else
                    plot_c_2=plot3(x,y,z,'or',MarkerSize=8);
                    hold on
                end
            elseif abs(S(2)-S(3))<0.05 && S(2)>S(1) % if b and c are tied
                if y>z % then use plurality to break tie
                    plot_b_2=plot3(x,y,z,'og',MarkerSize=8);
                    hold on
                else
                    plot_c_2=plot3(x,y,z,'or',MarkerSize=8);
                    hold on
                end
            elseif S(1)>S(2) && S(1)>S(3)
                plot_a=plot3(x,y,z,'ob',MarkerSize=8);
                hold on
            elseif S(2)>S(1) && S(2)>S(3)
```

```
                plot_b=plot3(x,y,z,'og',MarkerSize=8);
                hold on
            elseif S(3)>S(1) && S(3)>S(2)
                plot_c=plot3(x,y,z,'or',MarkerSize=8);
                hold on
            elseif abs(S(1)-S(2))<0.02 && abs(S(1)-S(3))<0.02
                plot_abc=plot3(x,y,z,'ok',MarkerSize=8);
                hold on
            end
        end
    end
end

axis equal
legend([plot_a,plot_b,plot_c,plot_abc], ...
    {'a','b','c','abc'})
xlabel('(a,b,c)')
ylabel('(b,c,a)')
zlabel('(c,a,b)')
title('Composed rule')
hold off
```

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